

SiViB Record 1500

SiViB Record Control

**Installation Instructions
and
Operation Manual**

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System Description



SiViB Record 1500 is a compact unit for complete machine condition monitoring. The most important condition indicators vibration, speed, temperature, and other process parameters are handled in only one monitor. Up to 370 data sets for all inputs and time stamp may be stored either in time intervals or based on trigger events like alarm conditions. The compact size and easy installation makes SiViB ideal for use on motors, pumps, fans etc. An input for selection between two different alarm sets allows the implementation on non continuously running machines, such as machine tools.

Functions:

Using the free configuration software, the ranges and alarm levels can be set easily and transferred from a PC to SiViB via the standard RS 232 serial link using a nullmodem cable. The configuration software runs under Windows 95, 98 and NT4.0.

During the operation of the machine, vibration is being monitored. When the vibration supersedes one of the alarm setpoints, the according output will switch to alarm.

If meaningful, an input controlling the active set of alarm setpoints can be used. The machine control system can provide the information if the machine is in idle or working under load. Different alarm settings might make sense and can therefore be applied using this feature.

The alarm outputs are optically isolated transistors, allowing an easy interfacing to a PLC System. LED's on the front panel indicate the alarm status.

The measurement values can be stored in an internal memory either in selected intervals or on trigger events, such as alarm conditions. The memory can hold up to 370 data sets each consisting of all input levels and a time stamp. The data can be transferred to a PC using the RS323 serial link or alternatively by a smartcard. The card is first inserted into the SiViB for being loaded with the data. Then it is taken to a PC with a smartcard reader connected. The software SiViB Record Control unloads the data from the card into the PC and stores to data files. A graphical trend display window in the software makes it easy to identify deterioration of machine condition.

The optionally available software package SiViB Record Control Pro adds features like frequency spectras and run up / coast down for further diagnosis and troubleshooting more complex machinery problems. Using this advanced software saves you the costs for an additional vibration analyzer.

Hardware Installation

Wiring Terminals for Type 1501, 1511

Terminal	Function
1	+ 24 V DC
2	Ground 24 VDC
3	Accelerometer 1 Signal
4	Accelerometer 1 Ground
5	Alarm A
6	Alarm B
7	Alarm C
8	Alarms Set Input (Set 1)
9	Machine Speed Input (1 pulse per rev)
10	+24 V DC Supply for alarm outputs
11	Ground alarms
12	Option Analog Out 0 – 10 Volt for mm/s for input 1
13	Option Analog Out 0 – 10 Volt for gSE for input 1
14	Ground for analog outputs
15	Ground for analog outputs
16 (1 st upper row)	AIN 1 (analogue input 1)
17	Analog GND
18	AIN 2.
19	Analog GND
20	AIN 3 or accelerometer 2 signal
21	Analog GND or accelerometer 2 ground
22	1501, 1511: AIN4 1502, 1512: Option Analog Out 0 – 10 Volt for mm/s input 2
23	1501, 1511: Analog GND 1502, 1512: Option Analog Out 0 – 10 Volt for gSE input 2
24	DIG IN 1 (digital input 1) / Event enable
25	DIG IN 2 (elapsed hour counter)
26	DIG IN 3 (tool change counter)
27	DIG IN 4 (store)
28	DIG GND
29	CAN Ser + (Option)
30	CAN Ser – (Option)

Description of terminal connections

- 1: +24 V DC supply
- 2: Ground supply
- 3: Accelerometer 1 Signal. For accelerometer with 4 mA constant current supply (ICP)
- 4: Accelerometer 1 ground
- 5: Alarm output A (Opto coupler output) max. 150 mA load (supply for opto couplers terminal 10 and 11 must be connected for use of all alarm outputs).
- 6: Alarm Output B (see above)
- 7: Alarm Output C (see above)

- 8: Alarm set input. Opto coupler input for 24 V DC signal from machine control system. Can be used to monitor two different operation conditions of the machine requiring different alarm level setpoints. Input = 1 (high) activates alarms set 1. If the terminal is not connected, alarms set 0 is always active.
- 9: Machine speed input. Opto coupler input for 24 V signal once per rev. Standard inductive non contacting pickups can be connected directly. Use terminal 11 as ground for this signal.
- 10: Supply for alarm outputs +24 V DC. Necessary for opto coupler outputs. Should be taken from the supply of the connected machine control system.
- 11: Ground alarms. Should be taken from the supply of the connected machine control system.
- 12: Analog output 0 – 10 Volts for mm/s overall of input 1. Use terminal 2 or 4 for ground.
- 13: Analog output 0 – 10 Volts for gSE overall of input 1. Use terminal 2 or 4 for ground.
- 14 – 15 Ground for analogue outputs
- 16: Analogue input 1
- 17: Analogue input 1 ground
- 18: Analogue input 2
- 19: Analogue input 2 ground
- 20: Analogue input 3; for 2 channel SiViB 1502 accelerometer 2 signal (like terminal 3)
- 21: Analogue input 3 ground; for 2 channel SiViB 1502 accelerometer 2 ground (like terminal 4)
- 22: SiViB 1501 and 1511: Analogue input 4;
SiViB 1502 and 1512 Analog output 0 – 10 Volts for mm/s overall of input 2
- 23: SiViB 1501 and 1511: Analogue input 4
.... SiViB 1502 and 1512 Analog output 0 – 10 Volts for gSE overall of input 2

Attention: The analogue inputs are preconfigured according to your order for 0 – 10 V DC or temperature sensors type PT1000, KTY84 or VT. These hardware configuration can not be changed by the end user. Please refer to the nameplate of the instrument, which states the configuration.

24 to 26: Digital inputs DigIn 1 to 3, opto coupler 24 V DC inputs for connection to PLC outputs. The status of these inputs get recorded with every other storage of vibration data.

DigIn 1 can get configured as "Event enable". If this is set, Events will only get stored if DigIn4 = high.

DigIn2 can be used as counter for operating hours

DigIn3 can be used as counter for tool change operations

27: Digital input „store“. A 24 V high signal causes the storage of a data set.

28: Digital Ground for digital inputs

29: Reserved for option CAN bus Ser+

30: Reserved for option CAN bus Ser-

Earting and Grounding: Depending on the earth situation it might be necessary to ground the instrument using one of the analog ground terminals (e.g. 4,17,19, 21). Please take care to use a "clean" earth carrying no leakage currents or other signals, as those would be seen as noise signals on the instruments inputs.

If the earth is heavily contaminated, it might be better to keep the monitor ungrounded and the sensor cable shield connected to signal ground only (terminal 4).

Where to mount the instrument: Choose a place for mounting the instrument, which is as away from sources of electrical noise. Motor drives, unshielded power cables or inductive transformers could be such sources, which should be avoided.

Wiring Terminals for Type 1502, 1512

Terminal	Function
1	+ 24 V DC
2	Ground 24 VDC
3	Accelerometer 1 Signal
4	Accelerometer 1 Ground
5	Alarm A
6	Alarm B
7	Alarm C
8	Alarms Set Input (Set 1)
9	Machine Speed Input (1 pulse per rev)
10	+24 V DC Supply for alarm outputs
11	Ground alarms
12	Option Analog Out 0 – 10 Volt for mm/s for input 1
13	Option Analog Out 0 – 10 Volt for gSE for input 1
14	Ground for analog outputs
15	Ground for analog outputs)
16 (1 st upper row)	AIN 1 (analogue input 1)
17	Analog GND
18	AIN 2.
19	Analog GND
20	Accelerometer 2 Signal
21	Accelerometer 2 Ground
22	Option Analog Out 0 – 10 Volt for mm/s for input 2
23	Option Analog Out 0 – 10 Volt for gSE for input 2
24	DIG IN 1 (digital input 1) / Event enable
25	DIG IN 2 (elapsed hour counter)
26	DIG IN 3 (tool change counter)
27	DIG IN 4 (store)
28	DIG GND
29	CAN Ser + (Option)
30	CAN Ser – (Option)

For further descriptions see above

Wiring Terminals for Type 1503, 1513

Terminal	Function
1	+ 24 V DC
2	Ground 24 VDC
3	Accelerometer 1 Signal
4	Accelerometer 1 Ground
5	Alarm A
6	Alarm B
7	Alarm C
8	Alarms Set Input (Set 1)
9	Machine Speed Input (1 pulse per rev)
10	+24 V DC Supply for alarm outputs
11	Ground alarms
12	Option Analog Out 0 – 10 Volt for mm/s for input 1
13	Option Analog Out 0 – 10 Volt for gSE for input 1
14	Ground for analog outputs)
15	Ground for analog outputs
16 (1 st upper row)	Option Analog Out 0 – 10 Volt for mm/s for input 2
17	Option Analog Out 0 – 10 Volt for gSE for input 2
18	Option Analog Out 0 – 10 Volt for mm/s for input 3
19	Option Analog Out 0 – 10 Volt for gSE for input 3
20	Accelerometer 2 Signal
21	Accelerometer 2 Ground
22	Accelerometer 3 Signal
23	Accelerometer 3 Ground
24	DIG IN 1 (digital input 1) / Event enable
25	DIG IN 2 (elapsed hour counter)
26	DIG IN 3 (tool change counter)
27	DIG IN 4 (store)
28	DIG GND
29	CAN Ser + (Option)
30	CAN Ser – (Option)

For further descriptions see above

Terminals of instruments for eddy current probes

(Type 1532 for 2 eddy current probes and 1 accelerometer)

Terminal	Function
1	+ 24 V DC
2	Ground 24 VDC
3	Accelerometer 1 Signal
4	Accelerometer 1 Ground
5	Alarm A
6	Alarm B
7	Alarm C
8	Alarms Set Input (Set 1)
9	Machine Speed Input (1 pulse per rev)
10	+24 V DC Supply for alarm outputs
11	Ground alarms
12	Option Analog Out 0 – 10 Volt for mm/s for input 1
13	Option Analog Out 0 – 10 Volt for gSE for input 1
14	Ground for analog outputs
15	Ground for analog outputs
16 (1 st upper row)	Eddy current probe 1 signal
17	Ground for eddy current probes
18	Eddy current probe 1 signal
19	Ground for eddy current probes
20	Supply for eddy current probes -24Volts max 40 mA
21	ECP1 Analog output 0-10 Volts DC for position
22	ECP1 Analog output 0-10 Volts DC for vibration
23	Out Ground (for analog outputs)
24	ECP1 Analog output 0-10 Volts DC for position
25	ECP1 Analog output 0-10 Volts DC for vibration
26	DIG IN 1 / Event enable
27	DIG IN 4 (store)
28	DIG GND
29	CAN Ser + (Option)
30	CAN Ser – (Option)

Description

Terminals 1 to 15 and 27 to 30 are equal to those on instruments for accelerometers. Please refer to the description above for further explanations.

16 and 18 Signal input for eddy current probes.

17 and 19 Ground for eddy current signal inputs. Attention: do not connect to other ground terminals!

20 Output for supply voltage -24 Volts, max. 40 mA for oscillator/demodulator (sometimes called proximitors or drivers). The corresponding ground terminals are 17 and 19. Attention: this supply is not short-circuit proof!

21 and 24 Analog outputs 0 – 10 Volts for the static (position) component of the probe signal.

22 and 25 Analog outputs 0 – 10 Volts for the dynamic (vibration) component of the probe signal.

Software Installation

PC Requirements

Minimal configuration for PC

Pentium Processor or compatible

64 MB RAM

HD min. 20 MB free

CD-ROM or DVD drive for software installation

Min. 1 RS232 serial port

VGA graphics, resolution min. 800 x 600; 256 colors

Operating System

SiViB Record Control runs under Windows 98, XP, Vista, Win7

Installing the Software

Put the CD into your CD-ROM drive.

Click Start, Run

Type:

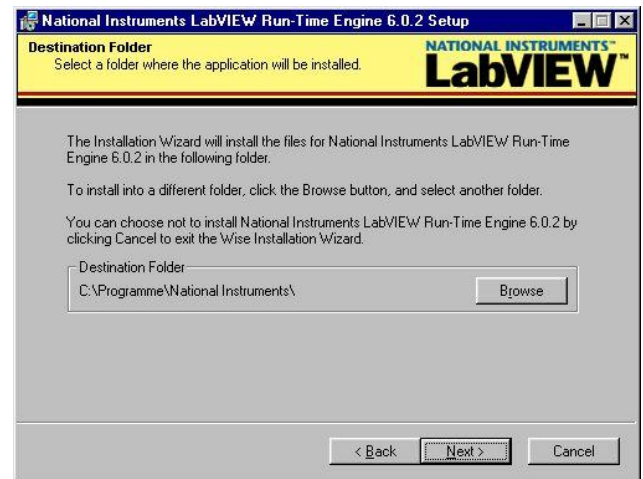
D:\English\setup (if your CD drive is not D:, use the corresponding drive letter instead).

Click OK.



The default path is C:\Record_Control. You can change the path according to your wishes, if required. Kick "Finish" for starting the installation process.

Firstly the Record Control program and configuration files will be installed. Then the „LabView Runtime Engine Setup“ will start automatically. The runtime engine is necessary for running Record Control. Follow the dialogue screens for completing the process.



This completes the installation process.

Reboot your PC for activating all new settings.

Install Chipcard Reader

The optional chip card reader has an 9-pin SubD connector for the COM Port and a PS/2 connector for powering the device. Both connectors must be connected to your PC. If you are using a Keyboard or mouse on the PS/2 port, this can be connected to the back side of the card reader connector. The driver of the card reader will find the reader automatically, so you do not need to set the port manually. There are other types of card readers with only the COM Port connector, and those for use on USB Ports. These types do not require additional power connections.

If you are using the card reader for Record Control only, you do not need to install any software that comes with the card reader. The Record Control installation installs all necessary drivers.

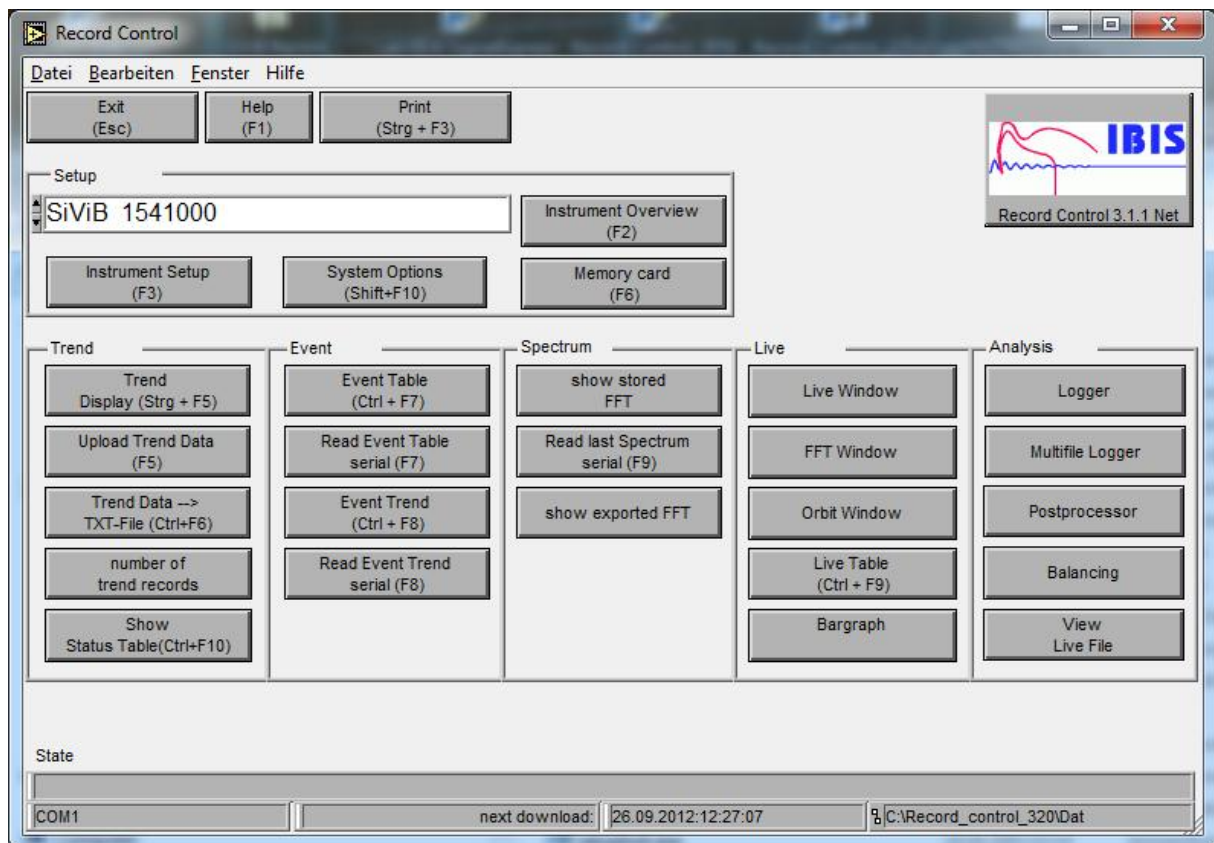
The card reader is an option. You can download data from SiViB Record without a card reader, using the RS 232 connection.

Attention: If you are using any of the software that comes with the card reader, make sure you close all these applications before you use the card reader for Record Control.

The chip card reader is not supported under Windows 7.

Starting the Software

For starting Record Control click Start, Record Control, Record Control



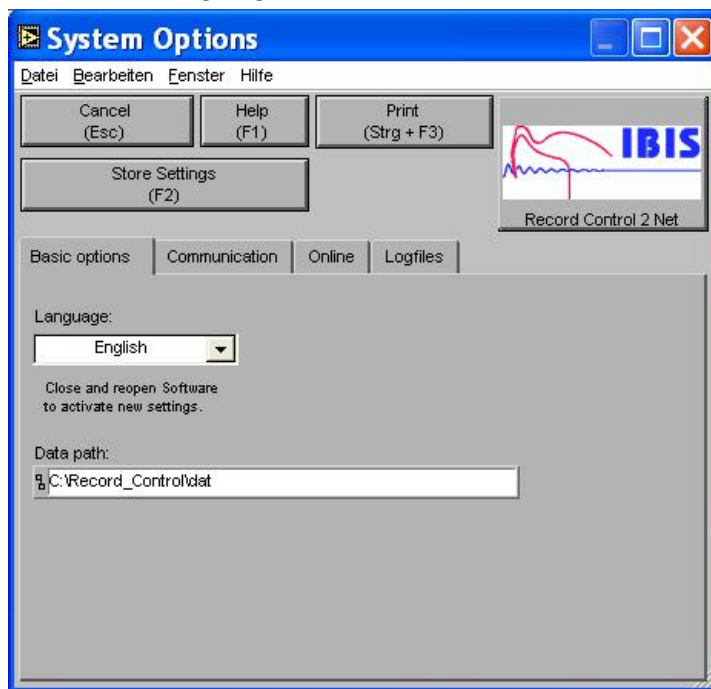
Basic Operation Principle

For quitting any open window in Record Control always use the "Quit" button or the Escape key of your keyboard. Avoid leaving windows open when jumping back. Otherwise windows in the background might stay active and the software appears blocked or misbehaving. Prefer clicking the "Quit" button in the upper left corner of most screens for closing a window. Some changes may not get stored if you click the "x" in the upper right corner.

If you switch to another software using the Windows task bar and come back to Record Control, the main window will appear on top. If you want to switch to another window in the background, just click on the Record Control button in the task bar repeatedly.

Basic Software Configuration

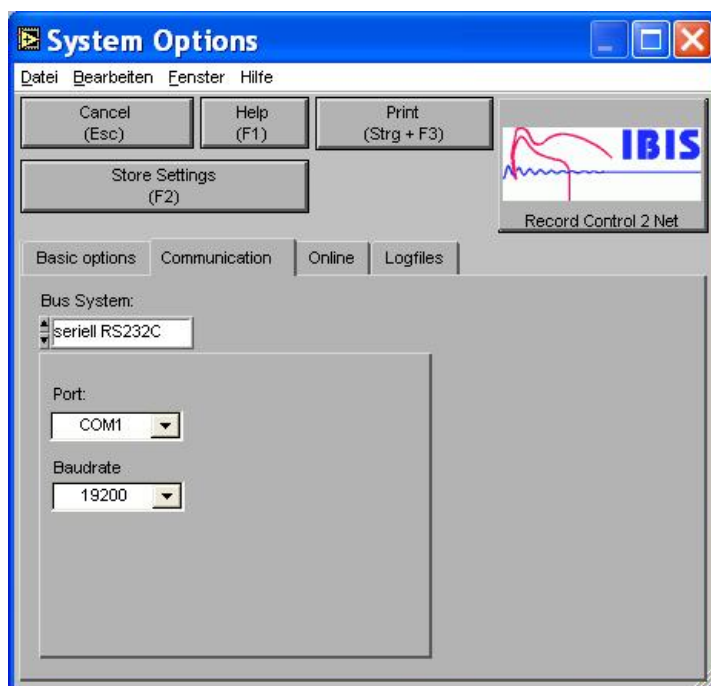
Software Language and Data Path



Select the language from the dropdown list box. Store the changes, quit and reopen the software to make changes active.

The standard data path is:
C:\Record_Control\Dat
You can use any other folder accessible by your PC, also if it is physically located on a server. Copy the files from the standard path to the required one and change the setting

Communication



Choose the bus system first:

RS232

Choose the COM port to use with a serial null modem (cross over) cable for communication between instrument and PC. The baudrate is 19200 baud.

Attention: If you are using one of the programs from the utility CD of the card reader, make sure you close it before opening Record Control.

The following options are only available in record Control Net:

CAN-Bus NI-CAN and P-CAN

For CAN-Bus adapters from National Instruments or Peak Elektronik
Each instrument must have a properly set CAN address. All nodes on the bus must be set to use the same baudrate.

Ethernet

For SiViB Records with Ethernet port.

Ethernet-CAN Gateway

For a cluster of SiViB Instruments and a Gateway module talking to the PC via Ethernet.

Multibus

For multiple monitors with Ethernet connector

Setup Procedure for SiViB Record 1500E

1. Set SiViB IP address

- Each SiViB has to be set to its individual IP address, which is stored inside the instrument..
- Connect SiViB and PC with an Ethernet cable. If you don't use a switch, a crossover cable has to be used for this operation.
- Power up the instrument
- Set your PC to the same address range as the standard address of SiViB.
The factory default IP address is 192.168.42.50, Subnet Mask 255.255.255.0
For the PC you can use any free IP address in the same range (e.g. 192.168.42.1)
Open your web browser and type the factory default IP address 192.168.42.50 into the web address field of the browser.
- A window asking for user name and password should appear. You can leave everything empty and just hit return.
- Go to the page **Network**

LANTRONIX®

Firmware Version: V6.1.0.0
MAC Address: 00-20-4A-8F-E5-7A

Network Settings

IP Configuration

☐ Obtain IP address automatically

Auto Configuration Methods

BOOTP: ☒ Enable ☐ Disable

DHCP: ☒ Enable ☐ Disable

AutoIP: ☒ Enable ☐ Disable

DHCP Host Name:

☒ Use the following IP configuration:

IP Address:

Subnet Mask:

Default Gateway:

Ethernet Configuration

☒ Auto Negotiate

Speed: ☒ 100 Mbps ☐ 10 Mbps

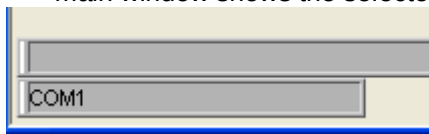
Duplex: ☒ Full ☐ Half

OK

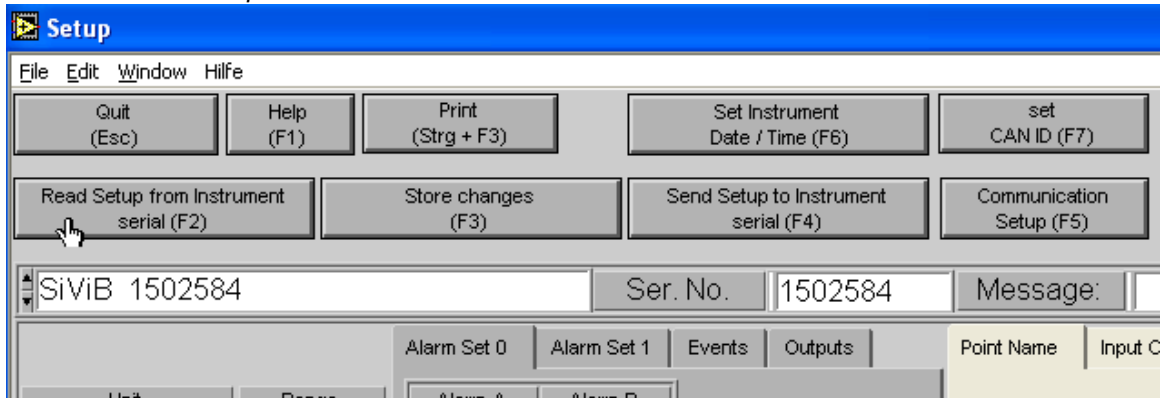
- Set the required IP address and Subnet Mask. A Gateway address is usually not necessary. Then click OK first, followed by Apply Settings. This will store these settings in a non volatile memory inside the instrument
- All settings on the other pages have already been set correctly during the production of SiViB. You should not change them.
- Set the PC with Record Control software to a free IP address in the same address range.

2. Register SiViB in the software

- Connect SiViB to the PC using a nullmodem RS232 cable.
- Make sure the software is set to serial RS232 communication. The status line at the bottom of the main window shows the selected COM Port number.

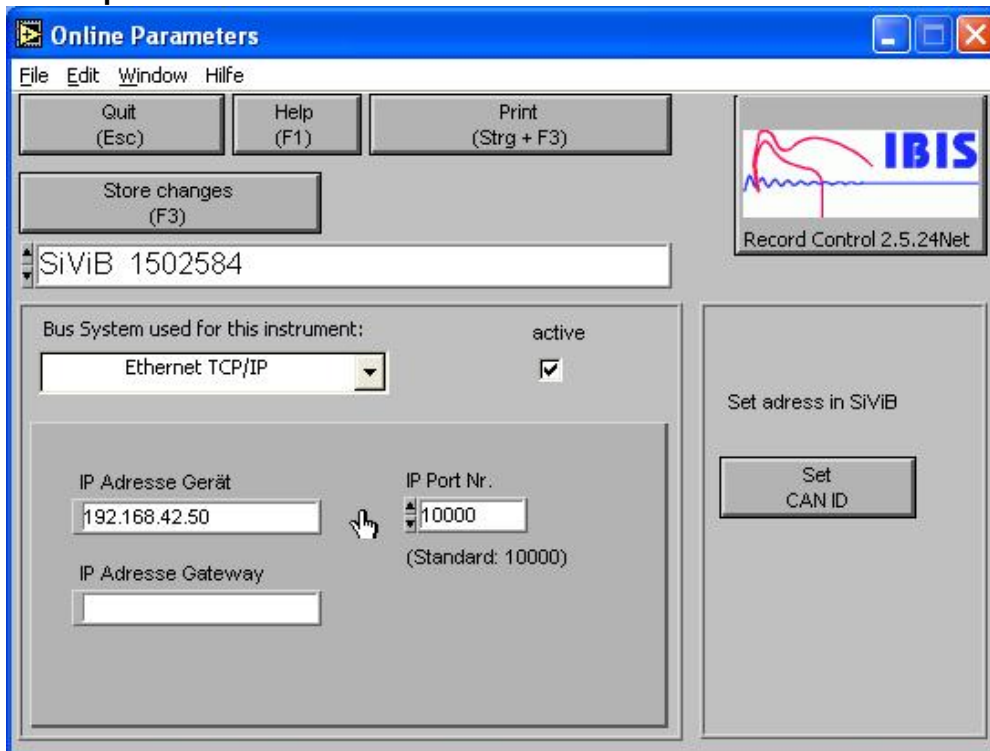


- Go to the Setup Window
- Click *Read Setup from Instrument*



- Allow the system 3 – 5 seconds to transfer all settings to the PC
The Instrument's name and serial number should appear in the corresponding fields.
- Click Store changes to write the setup to the files in the PC

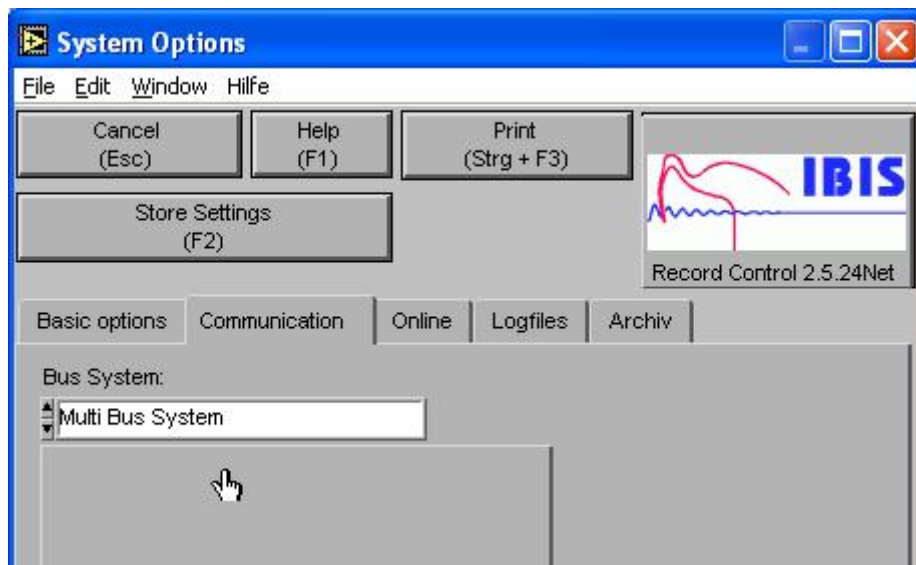
3. Setup the communication details for this SIVIB



- Type in IP adress and Port, check the active flag and click **Store changes**
This stores this settings in the online parameters file of the software

4. Set Global Communication Options

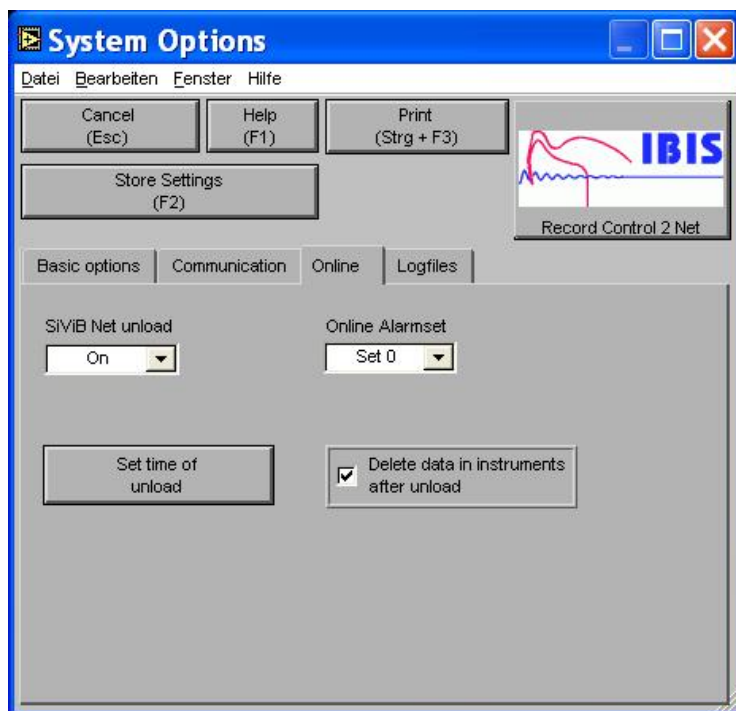
- Goto **System Options** (from Main Window), Tab Communication



- Select Multi Bus System, click Store Settings. Password is **sivib**
Close Record Control and reopen it. This makes the changes in system options active.
Please note that the communication port(s) to the SiViB(s) are opened when Record Control is launched.

Note: If you have more than one SiViB you will have to repeat steps 1 to 3 for each SiViB. First do step 1 for all instruments, then step 2 for all and so on. Step 4 is common for all instruments.

Online (Record Control Net only)



Net Unload:

Automatic upload of trend data from the instruments to the PC.

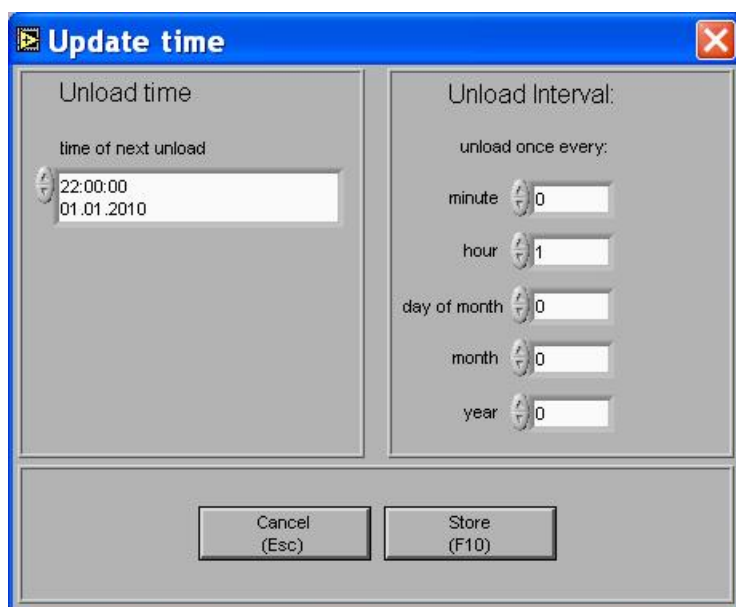
Online Alarmset:

In the system overview table, points in alarm are set to yellow or red. This setting chooses which alarm set to use for the overview table.

Delete data in instrument after unload. Deletes the data memory of the instrument after automatic unload. If you are using "email on alarm", this avoids repeated alarming of old alarm conditions.

Set time of unload

Opens a window for setting the time and interval of automatic unloads.

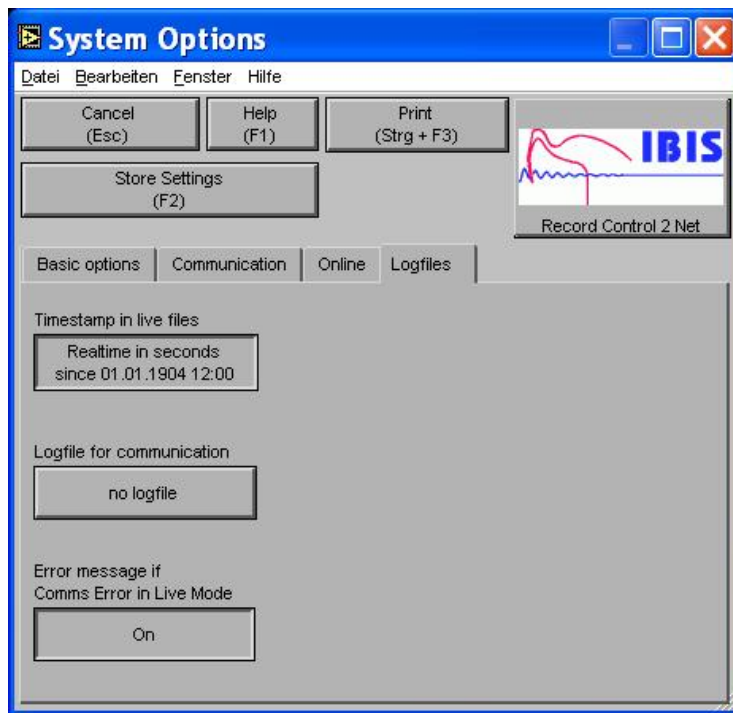


Under „time of next unload“, set the time and date of the next automatic trend data transfer from the instrument(s) to the PC.
hour:minute:second
day.month.year

The unload interval specifies the time interval between automatic unload operations.

After each unload procedure, the interval will be added to the unload time and stored as new next unload time.

Logfiles



Timestamp in live files

Realtime in seconds since 01.01.1904, 12:00 or relative time in seconds since start of recording. Please find more details on this option in the chapter describing store to file option in live mode.

Logfile for communication

If set to "write logfile", all communication telegrams sent between instrument and PC get logged to a file. Attention: The log file will get very large if you keep it running for a longer time. Use this option for troubleshooting only.

Error message if comms error in live mode.

On: In the case of communication errors a popup window showing an error message will appear and wait for the user to confirm by clicking OK.

Off: No error message in the case of comms problems.

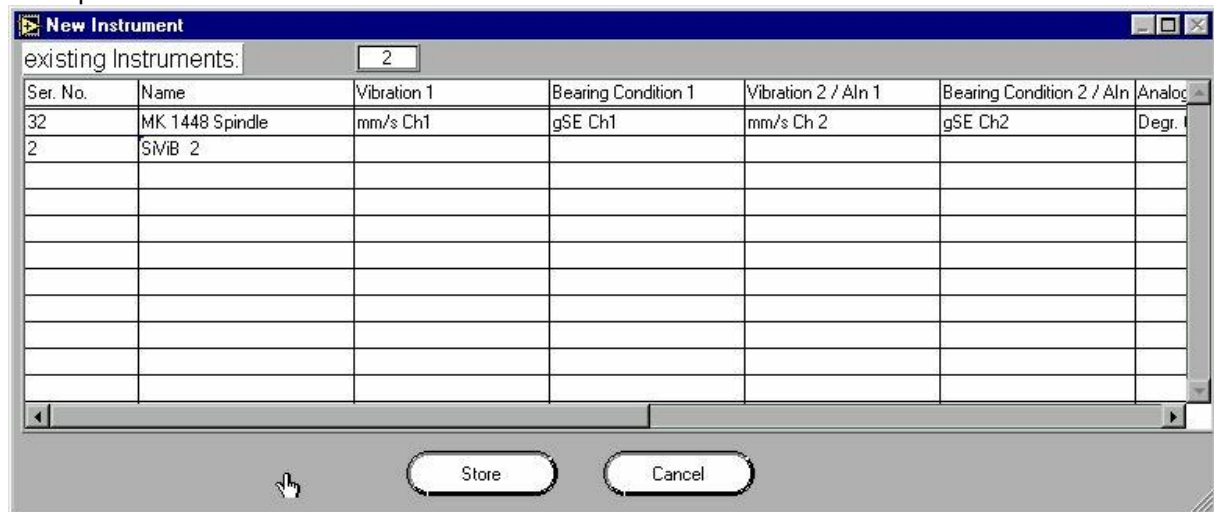
Attention: If you plan to use the write to file option in life mode for a longer time without user attendance, it is recommended to set this option to off.

Configuration and Commissioning

New Instrument

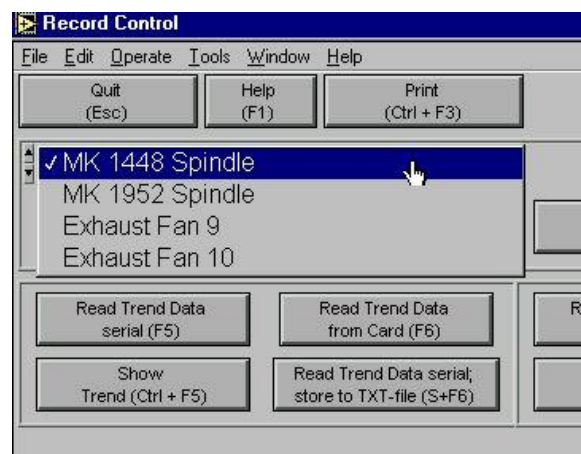
For commissioning a new SiViB connect the instrument to your PC using a null modem cable. Connect the 24 Volts power supply and switch it on. Boot your PC and start Record Control. From main window select **New Instrument**. Fill the next empty row of the table with the serial number (see nameplate) and the names for the measurement points according to your application.

Example:



Click **Store** for saving to file and quitting the window.

Now you will find the instrument's name in the selection box in the main window.



Select the required instrument.

Instrument Setup

The screenshot shows the 'Setup' window of the SiViB Record 1500 software. The interface includes a menu bar (Datei, Bearbeiten, Fenster, Hilfe) and a toolbar with buttons for Quit (Esc), Help (F1), Print (Strg + F3), Set Instrument Date / Time (F6), set CAN ID (F7), Read Setup from Instrument serial (F2), Store changes (F3), Send Setup to Instrument serial (F4), and Set time of unload (F8). The main area displays 'SiViB 153217' and 'Ser. No. 153217'. It features tabs for Alarm Set 0, Alarm Set 1, Events, and Outputs. A table lists units and ranges for various measurements: Vibration mm/s (0 - 10), Bearing condition (0 - 10), Position µm (0 - 1000), Vibration µm (0 - 100), Position µm (0 - 1000), Vibration µm (0 - 100), and RPM x 1000 (0 - 60000). There are also input options for Point name and Input options. The bottom section contains 'Storage mode' settings (Store input, Trend storage interval, Interval FFT, Sample Rate, Interval [s]) and a 'Settings Database' table with 7 rows (Erste to Siebte).

Click **Setup, Read Setup from Instrument serial**

Select all settings according to your requirements. The different settings have the following meanings:

Unit

Depending on sensor type and input, different units of measurement are selectable. If you are using accelerometers, you can select vibration [mm/s RMS] and bearing condition units Spike Energy [gSE]. Velocity is always the first channel, bearing condition the second. Channels 3 and 4 are dependent on the type of SiViB you are using. SiViB 1501 and 1511 have analogue inputs in line 3 to 6, which could be used for 0 – 10 V DC or temperature. The type of input is defined by the hardware. Refer to the nameplate of the instrument and set the input configuration accordingly. SiViB 1502 and 1512 both have a second input for an accelerometer. Set line 3 for vibration [mm/s] and line 4 for bearing condition [gSE]. On these type of instruments analogue input 4 is not connected.

Range

Select the required measurement range. Please remember that the values of your reading could be significantly above normal level in the case of developing defects.

Analogue voltage input is always 0 – 10 V DC.

For temperature sensors, select the sensor type (see nameplate). The range for temperature is 0 – 200 degrees Celsius.

Alarm Set 0

SiViB has 3 alarm outputs A, B and C. You can set an alarm level for each output. The output will change state if one of the corresponding alarm levels is superseded. For example, use alarm A for warning and alarm B for shutdown. All alarms can be set over the whole measurement range. The alarms of the different inputs and units are OR wired to the outputs. This means any of the units could switch the output when superseded. If you want an output to be effected by only one unit, set the alarm levels of all other units to maximum for this colon.

Alarms Set 1

The alarms set 1 are activated when terminal 8 goes high (24 Volts). This could be used to set different alarm levels for different operating conditions of the machine (idle / working or low speed / high speed).

Events

The event level sets the trigger level for an event. Besides this, it switches alarm output C. Event Trend Interval sets the interval time for event trend recording (see description below). You can set different Event levels for alarm set 0 and set 1 (in the columns below the corresponding tabs).

The following option “alarm active” is also important for the activation of event storage. Set all inputs to active which should trigger events.

The button “Event recording always on” or “Event recording only if DigIn1 = high” allows you to make event recording dependant on the status of the digital input number 1. Use this option to prevent false event alarms for operation conditions where it is normal for the monitored machine to generate high signals.

Outputs

Alarm active

Defines if the alarm outputs shall be switched by this unit / input. Activates the Event capture functions described above.

Alarm delay

Defines for how long time an alarm condition must be present, before the corresponding alarm output switches.

Alarm delay C delays the event recording too.

Alarm mode

Always = alarm output switches always if conditions mentioned above are met.

Only on Set 1 = alarm output switches only if conditions are met AND alarm set 1 is active.

Output level

Defines if the alarm outputs should be high (1) on alarm or low (0) on alarm. Using “0 on alarm” gives you the possibility to check for broken wires.

Input Options:

Point name	Input options
Averages	Sensitivity %
Offset	
10	100
10	100
10	100
10	100
10	100
10	100
10	100
0	0
0	0
0	0
0	0
32	32
32	32
Calibrate NCPUs	

Averages

Number of single overall values SiViB takes for calculating an averaged value before further processing. If you have fluctuating values, try to increase the number of averages for getting more stable readings.

Sensitivity

Pickup sensitivity. 100 corresponds to the standard input sensitivity of the input. Use this for correction of sensitivity deviations of the particular sensor connected.

Example: A particular sensor has a nominal sensitivity of 100 mV/g, but the sensor you got has only 95 mV/g. Set the sensitivity to 95 and the software will calculate the exact true value for you.

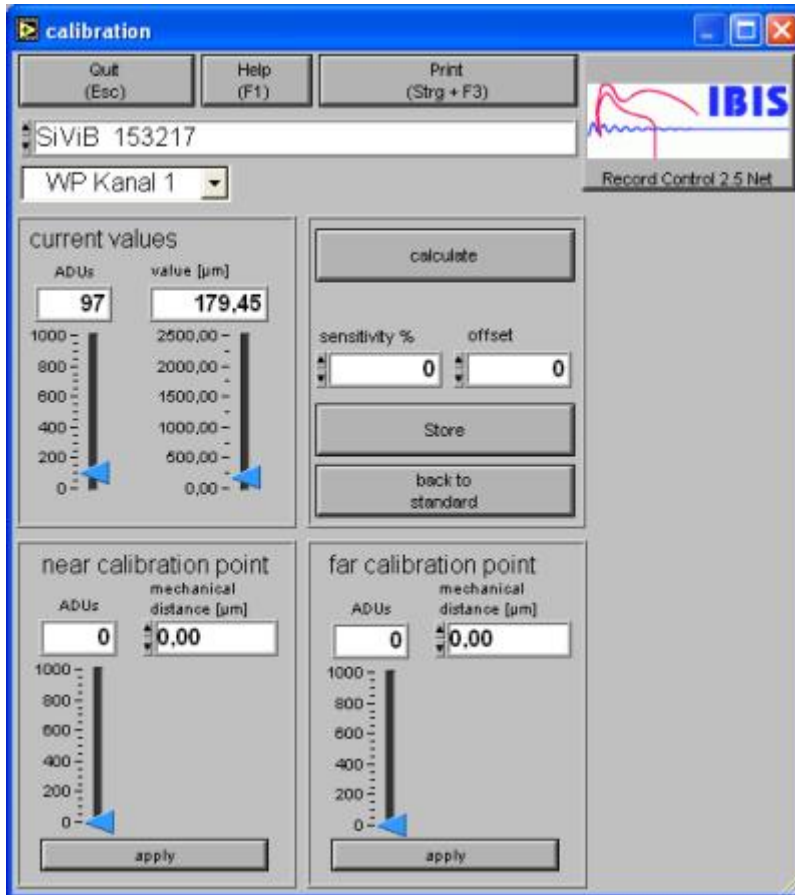
Attention: SiViB Record 1511 and 1512 have an input sensitivity of 10 mV/g for accelerometers. This means 10 mV/g is 100% sensitivity. A sensor with 9.8 mV/g would need a setting of 98 in the sensitivity field.

Offset

The offset value is added to the measured value of the channel. Negative offsets are allowed as well.

Calibrate NCPUs:

Standard calibration for the NCPU inputs of SiViB Record Type 1532 is 8 mV/ μm . As the sensitivity of NCPUs depends on the target material, it might be necessary to calibrate the inputs for getting true values. The calibration window helps to find the correct sensitivity and offset values.



Calibration procedure:

Move the sensor to the near calibration point close to the target (recommended 100 to 200 ADUs current value)

Measure the mechanical distance between sensor tip and target, enter the value into the field and click apply

Move the sensor to the far calibration point (recommended 800 to 900 ADUs current value)

Measure the mechanical distance between sensor tip and target, enter the value into the field and click apply

Click "calculate". Now the software shows true values in the block current values.

Click "store" to write the calibration for this input channel to the setup file.

Repeat the procedure for the second input channel, if you use it too.

Storage Mode

Defines under which conditions data sets shall be stored to the trend data memory.

Interval timer: system stores a data set to trend memory after each time interval.

Store input: system stores a data set to trend memory each time the store input (terminal 27) gets high.

Interval and store input: system stores a data set to trend memory on the next store input signal after the time interval has expired.

Interval or store input: system stores a data set to trend memory after interval time has expired or if the store input goes high.

Within speed range, Interval: system stores a data set to trend memory as soon as the speed window is reached, and then after each interval period, as long as within speed range. The speed window is defined by the two speed alarm settings on tab Alarm Set 0.

Within speed range, after interval: after reaching the speed window the system waits for interval time before the first data set is stored. Then it stores after each time interval as long as the speed window is true.

Interval FFT,

The settings for Interval Spectrum are only relevant for users of Record Control Pro software. It defines in which intervals and at which sampling rate a time waveform should get stored for FFT spectrum calculation. The instrument can hold only the most recent spectrum (respectively time waveform) for each vibration input. If the interval is set to 0, the system will not take spectral readings.

Setting Database

No.	Setup:
1	Mill 234
2	Fan
3	Pump
4	Test stand type 1
5	not used
6	not used
7	not used

The settings database allows you to store data sets of frequently used setups. This makes it easy to configure multiple instruments using the same parameters. There are 99 data sets available.

How to store a configuration in the settings database:

1. Make all required entries in the options described above. Click "store changes".
2. Choose a data record in the settings database by clicking on the desired row. Enter a name for the data set.
3. Click "Store setup to file". Type your password and click OK. The data set is stored now.

How to use a configuration set from the setting database:

1. Select the instrument which shall get configured from the instrument selection dropdown box.
2. Select the configuration set from the settings database by clicking into the desired row.

3. Click "Read setup from file". The settings from the settings database will replace the current settings in the upper part of the window.
4. Click "Store changes" to make the settings valid.

Send Setup to SiViB

Important:

Store the settings by clicking "Store Settings" before you send them to SiViB by clicking on "Send settings to Instrument" The software sends only stored settings correctly.

Be aware that you need to send changed settings to the instrument before they get active.

Important:

Sending a new setup will erase all data stored in the instrument, as otherwise they could get inconsistent.

Set SiViB Date / Time

For adjusting the internal real time clock you must connect to the instrument using a null modem cable. Click **Setup, Set instrument date / time**.



The window shows both date and time from the PC and the instrument.

Click **Send data / time to instrument** for synchronizing the instrument's real time clock.

As the real time clock is battery powered, it will continue to run even when the instrument is power off.

Deleting data records in the instrument

If you want to delete any unvalid data records in the instrument, you can do this by sending the configuration. This will reset all data memory blocks.

Operation of the System

SiViB Record stores all settings in a battery powered memory, so it will wake up on power on and start with normal operation automatically.

During operation, SiViB monitors all inputs and compares the values against the alarm levels. In case of alarms the outputs will be switched. The alarm outputs may be connected directly to the machine control system, which could give warnings on the operator panel, or shut down if required.

The alarm LEDs on the front panel indicate the alarm state.

SiViB will store data records of the overall values according to the time interval or on external signal applied to digital input 4. The trend memory holds a maximum of 370 data records in fifo (first in first out) mode. Each record holds all overall values, digital input state and date/time stamp.

Transfer of Data Records to PC

You can upload the data records to your PC in two different ways.

1. Via RS232 serial link, CAN-Bus or Ethernet
2. Via a memory card (chip card). This option might be handy if you do not want to carry your PC to the machine, where your SiViB is installed.

Upload Data via RS232 / CAN-Bus / Ethernet

Connect your PC to SiViB using a null modem cable. Use the COM port which is configured in settings.ini (see installation instructions above). Alternatively, use CAN-Bus or Ethernet connection. Start the Record Control software.

In the main screen, click **Read Trend Data serial**.

Watch the status bar at the bottom of the screen for messages telling you the progress of the data transfer. Record Control checks the serial number of the instrument and if there are already data files for this instrument existing on the PC. If the files exist, the new data sets will be appended. If the files do not yet exist, Record Control will create them and use the name "SiViB XXXX" for the instrument, where XXXX stands for the instruments serial number. You can edit the name later using the screen

New Instrument.

For uploading the event memory content to the PC click **Read Event Table serial** or **Read Event Trend serial**.

If you own Record Control Pro, you can upload the most recent spectrum (interval spectrum) by clicking **Read last Spectrum serial**.

Upload Data via Memory Card

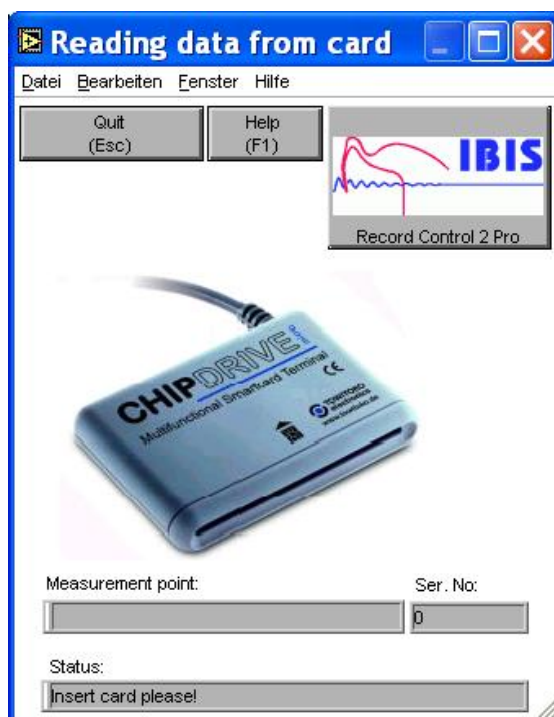
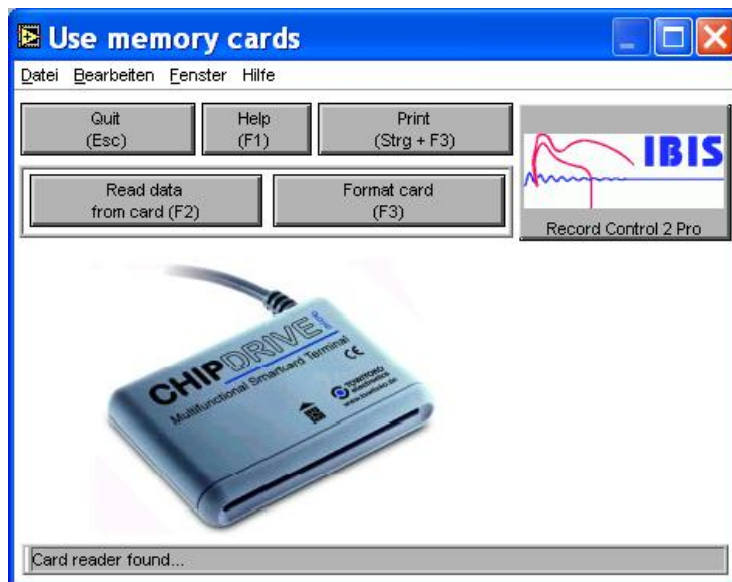
For uploading measurement data using a memory card you need a "formatted" chipcard. If your chipcard is not formatted yet, click **Read Trend Data from Card** and then **Format Card** (details below).

Plug the chipcard into the SiViB card reader slot with the golden contacts downwards (towards the RS232 connector). If the card was plugged in wrong side up, the red LED „Card Error“ will go on. If the card does not belong to this instrument (wrong number), the red LED will also go on. If everything is OK, the green LED „Card OK“ stays on as long as the data transfer is working. When the green LED goes off, you can remove the card. Depending on the amount of data, the data transfer could last up to 1 minute.

Attention: Do not remove the card as long as the green LED "Card OK" is on! Otherwise the data on the card could get incomplete and corrupted!

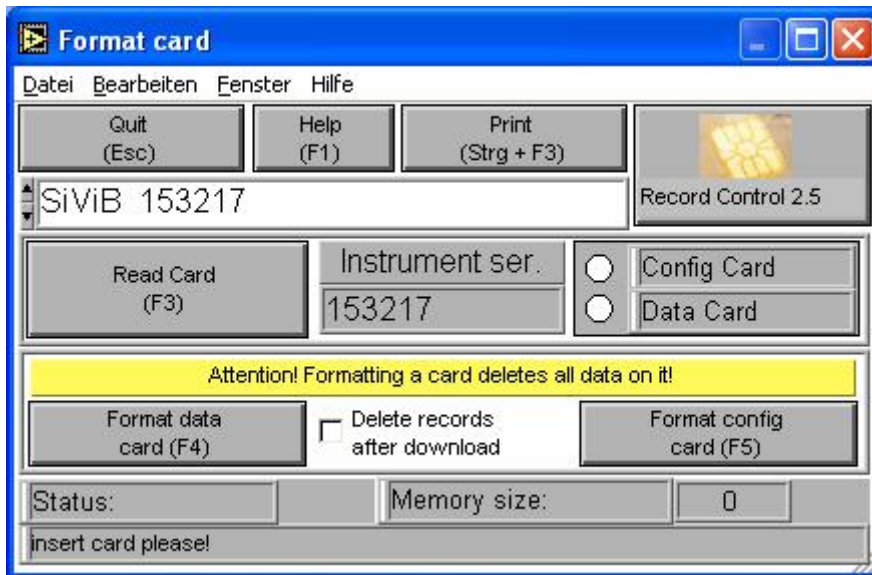
The upload process via memory card always transfers all data (trend, event table, event trend, spectrum).

Back at your PC click **Memory Card** in the main screen and **Read data from card**:



Wait until the status line says "insert card please". Plug the card into the reader with the golden contacts to the upper side. The next message will tell you which instrument the data belongs to and asks if it shall be stored in the files. When the data upload is finished, you will be asked if you want to proceed with another card.

Formatting Memory Cards



For making a new memory card usable for SiViB it must get “formatted” first. This will, among others, write the SiViB serial number to the card. This makes it impossible to swap data erroneously.

Attention! Formatting a memory card deletes all data on it irreversibly. There is no “undo” for format!

Connect your card reader according to the installation instructions above.

In the main screen click **Read Data from Card**. Then click **Format Card**. Plug the new card into the card reader. In the status window, you should see a message “Card recognized. Enter serial number and click Format”

Select the instrument, which you want to use the card for, using the instruments selection drop down box. The serial number shown on the instrument’s nameplate (just the number without any slashes or letters other then numbers) will then be shown in the serial number display box.

You can format **config cards** or **data cards**.

Config cards are used to transfer setup information from a PC to the instrument. During the format process, all settings are transferred and stored onto the card. When you plug the card into the instrument, the setup parameters are being transferred to the instrument and stored in its non volatile memory.

Data cards are used to transfer measurement data from the instrument to the PC. After formatting a card as data card, plug it into the instrument to transfer the data from the internal memory to the card.

After clicking either **Format Config Card** or **Format Data Card**, please wait for the finished message in the status line.

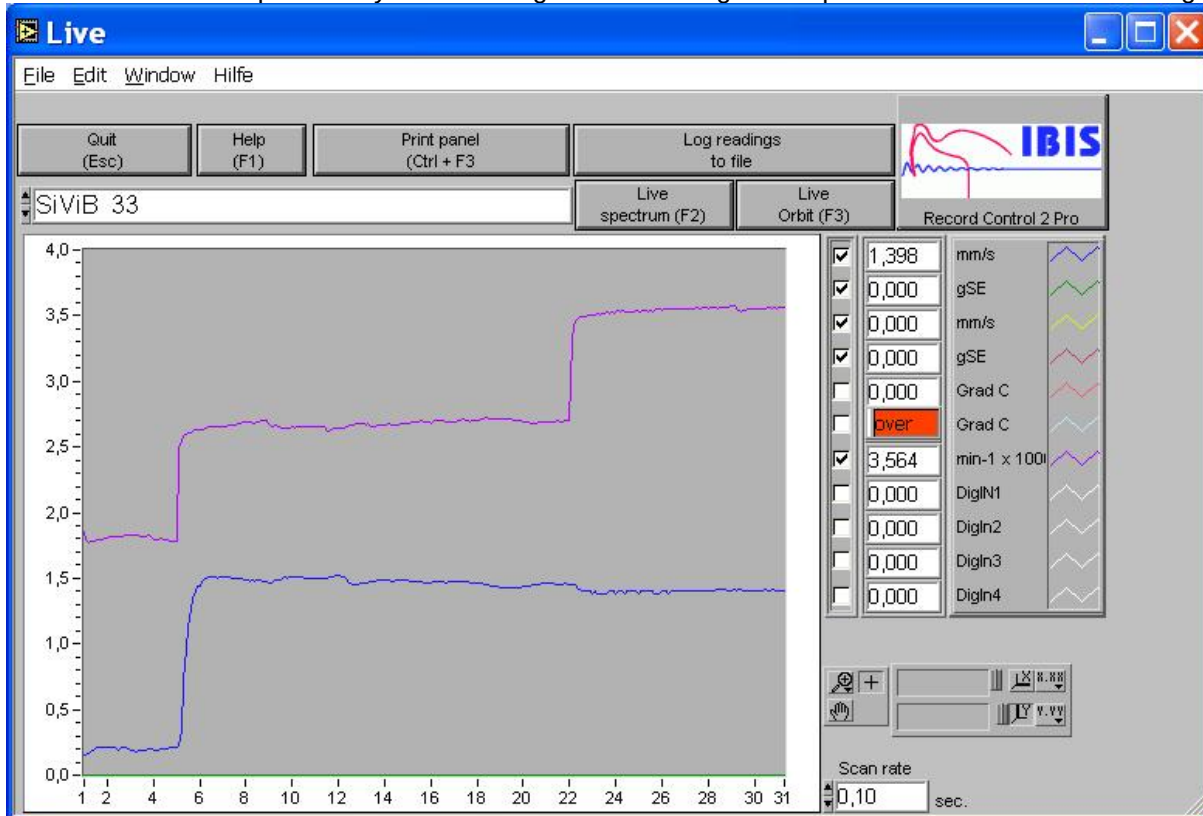
We would recommend writing the serial number onto the card for easy identification.

You can read the type and serial number of a card using the button **read card**.

You can reuse cards for other instruments or as other card type after reformatting them according to your needs.

Live Display

The live display allows you to watch the measurement data almost at the same time as they get measured from the instrument. A graphical chart shows the values versus time. Use this function for watching changing vibrations e.g. during startup or coastdown of a machine or during the cycle of a machine tool. This is particularly useful during commissioning or for special tests and troubleshooting.



Connect your PC to the instrument using the COM Port and null modem cable.

From main screen click **Live Display**.

The checkboxes between the graph and the digital display activate the curve for the different units on the plot.

The control **Sample Rate** sets the time interval between two values to be uploaded.

The switch **Graph** activates or deactivates the chart. On slow PCs this can be useful when writing to file.

When a signal is higher than the measurement range, a red "over" indicator will replace the digital value. You might need to go to Setup and change the measurement range.

Storing Data in Live Mode

If you click **Log Readings to File** all data transferred from the instrument will get stored to file. A dialogue box allows you to enter the file name. Writing starts after you have entered the file name. If you switch off writing, the logging process gets paused. If you switch writing on again, it will continue adding data to the same file. If you quit live mode and come back again and start writing, you will be prompted for a file name again. If you enter the name of an existing file, new data will get appended. The format of the logfile is tab separated ASCII text. Many software packages like spreadsheets will easily import files of this type.

The file format is as follows:

Colon:	Content:
1	Seconds since start of recording (or seconds since 01.01.1904 12:00)
2	Vibration velocity in [mm/s RMS]
3	Bearing condition in [gSE]
4	Analog 1 (or velocity channel 2)

5	Analog 2 (or bearing condition channel 2)
6	Analog 3
7	Analog 4
8	Machine speed in RPM [min ⁻¹]
9	Status Digital inputs (Bit wise)

You can get a reference to real time looking at date and time of the file using Windows Explorer. It gets updated on every write.

Alternatively, the software can write the timestamp as realtime (seconds since 01.01.1904 12:00). To select this option, go to **system settings, logfiles** and set the switch timestamp in live files to **seconds since 1904**.

Converting seconds since 1904 to time and date string

If you want to use the live data files with a spread sheet software (e.g. Excel), you can convert the seconds since 1904 to a time and date string using the following formula:

$((\text{seconds}/60/60+\text{timezone})/24+4*365+2)$
Format this cell as date

Timezone is the difference between universal time (GMT) to your local time.

Remark:

Excel has an option to use a different timebase:

Go to Extra, Options, Calculations, select "1904 timebase"

If you use this option, leave away the last part of the formula (+4*365+2).

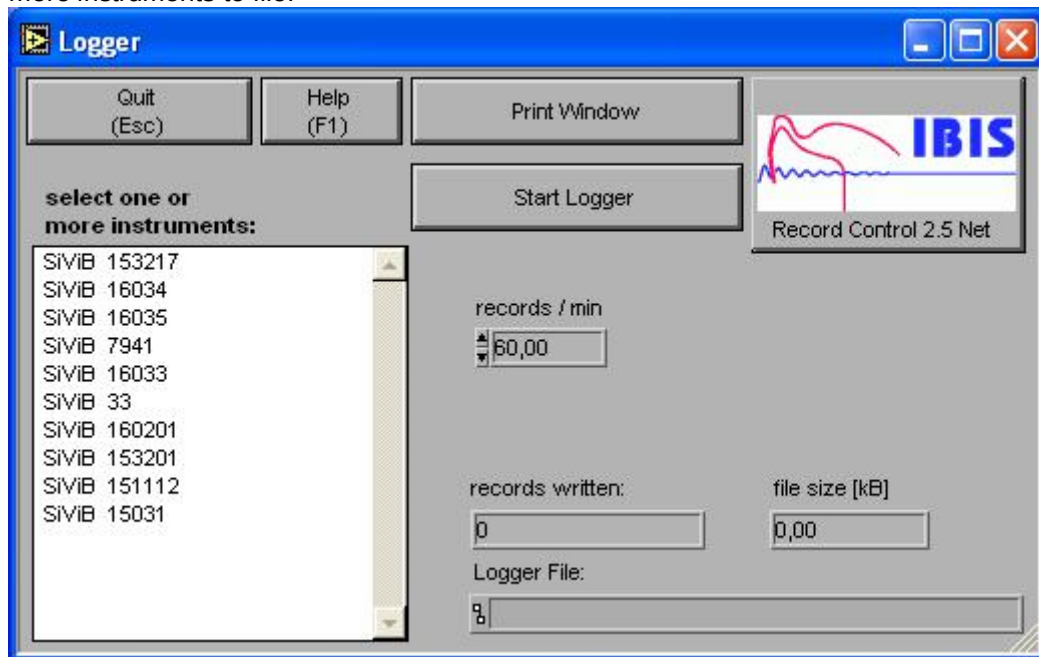
Example:

1. Open the file with Excel
2. Mark the first colon and insert an empty colon.
3. In the new colon, type in the formula stated above and copy it to the other lines.
4. Format the new colon as date or time according to your needs.

	A	B	C	D	E
1	10.9.02 14:05:45	3114507945,26	0	0	
2	10.9.02 14:05:56	3114507956,24	0	0	
3	10.9.02 14:05:56	3114507956,46	0	0	
4	10.9.02 14:05:57	3114507956,68	0	0	
5	10.9.02 14:05:57	3114507956,84	0	0	
6	10.9.02 14:05:57	3114507957,06	0	0	
7	10.9.02 14:05:57	3114507957,28	0	0	
8	10.9.02 14:05:57	3114507957,45	0	0	
9	10.9.02 14:05:58	3114507957,67	0	0	
10	10.9.02 14:05:58	3114507957,89	0	0	

Log to File

The Software version Pro and Net have a data logger function for writing overall values from one or more instruments to file.



Select one or more (only in Net) instruments and click "start Logger". You will be prompted to enter a file name and logging starts after you have completed the entry.

The contents of the logfile is similar to the files written from the Live window. If you have selected more than one instrument, the table will have additional colons for the values. Refer to the description above for more details of the file format.

The advantage of the Log to file function against writing from the Live window is that you can use it beside the other functions of the software. You can minimize the logger window and work with the other functions of the software without interrupting the logger. The only exceptions are Life FFT and orbit. During the collection of the time waveworms for FFT and orbit, the instrument can not answer the logger's request for data.

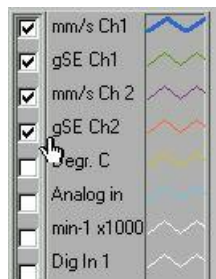
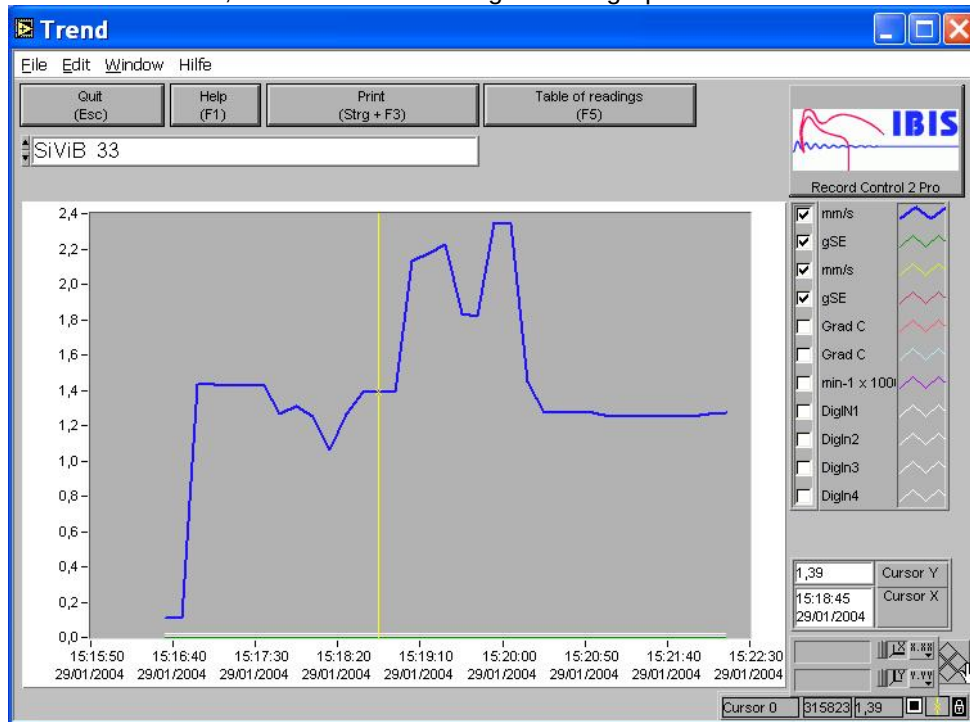
Reviewing and detailed inspection of files created by the logger is usually done using Microsoft Excel or other 3rd party software.

Alternatively you can use the function "View Live File" for reviewing and working with the zoom and cursor options.

Tend Graph

The trend graph is useful for looking at longer term changes in the values, maybe pointing at deterioration of the machine. Trend graphs are based on data taken regularly under comparable conditions. The trend memory of the instrument can hold up to 370 data records. Upload the data either using serial cable or the memory card (see above).

From main screen, click **Show Trend** to get to the graph.



Activate the curve(s) for the data of interest by checking the box.

Trend Table

From trend graph screen, click **Table of Readings** to show a numerical overview table.

Date/Time	mm/s	gSE	Analog1	Analog2	Analog3	Analog4	RPM	Dig In	Trigger
29.01.2004 15:16:35	0,111	0,000	0,000	0,000	23,262	212,726	1906,00	0,000	0,000
29.01.2004 15:16:40	0,111	0,000	0,000	0,000	23,262	212,726	1908,00	0,000	0,000
29.01.2004 15:16:45	1,435	0,000	0,000	0,000	23,553	212,726	2743,00	0,000	0,000
29.01.2004 15:17:00	1,435	0,000	0,000	0,000	23,262	212,726	2755,00	0,000	0,000
29.01.2004 15:17:15	1,423	0,000	0,000	0,000	23,262	212,726	2771,00	0,000	0,000
29.01.2004 15:17:30	1,423	0,000	0,000	0,000	23,553	212,726	2765,00	0,000	0,000
29.01.2004 15:17:45	1,423	0,000	0,000	0,000	23,262	212,726	2777,00	0,000	0,000
29.01.2004 15:17:50	1,262	0,000	0,000	0,000	21,193	212,726	2763,00	0,000	0,000
29.01.2004 15:17:55	1,312	0,000	0,000	0,000	20,302	212,726	2786,00	0,000	0,000
29.01.2004 15:18:00	1,250	0,000	0,000	0,000	20,005	212,726	2783,00	0,000	0,000
29.01.2004 15:18:15	1,064	0,000	0,000	0,000	20,005	212,726	2796,00	0,000	0,000
29.01.2004 15:18:30	1,262	0,000	0,000	0,000	20,005	212,726	2761,00	0,000	0,000
29.01.2004 15:18:45	1,386	0,000	0,000	0,000	20,005	212,726	3626,00	0,000	0,000
29.01.2004 15:18:50	1,386	0,000	0,000	0,000	20,005	212,726	3630,00	0,000	0,000
29.01.2004 15:18:55	1,386	0,000	0,000	0,000	20,005	212,726	3630,00	0,000	0,000
29.01.2004 15:19:00	2,129	0,000	0,000	0,000	20,005	212,726	4469,00	0,000	0,000
29.01.2004 15:19:15	2,178	0,000	0,000	0,000	20,005	212,726	4507,00	0,000	0,000
29.01.2004 15:19:30	2,228	0,000	0,000	0,000	20,005	212,726	4528,00	0,000	0,000
29.01.2004 15:19:45	1,832	0,000	0,000	0,000	20,005	212,726	6208,00	0,000	0,000
29.01.2004 15:19:50	1,819	0,000	0,000	0,000	20,005	212,726	6281,00	0,000	0,000
29.01.2004 15:19:55	2,351	0,000	0,000	0,000	20,005	212,726	4589,00	0,000	0,000
29.01.2004 15:20:02	3,39	0,000	0,000	0,000	20,005	212,726	4573,00	0,000	0,000

Depending on the user's rights you can delete rows of this table. Mark the line (drag the mouse while left button pressed), then click right and select cut data.

Click **Store Changes** and type your password

Store Trend Data to TXT File

If you want to process the trend data using other software, you can transfer the data records from the instruments trend data memory directly to an ASCII (TXT) file. From main screen click **Read Trend Data serial, Store to TXT File**. Watch the status line for progress messages and enter a file name when prompted.

The file format is tab separated ASCII (TXT).

The file format is as follows:

Colon:	Content:
1	Date and Time
2	Vibration velocity in [mm/s RMS]
3	Bearing condition in [gSE]
4	Analog 1 (or velocity channel 2)
5	Analog 2 (or bearing condition channel 2)
6	Analog 3
7	Analog 4
8	Machine speed in RPM [min^{-1}]
9	Status Digital inputs (Bit wise)

Event Memory

A “event” is defined as a special alarm situation which shall be stored separately. SiViB has two event memory blocks with different use:

1. Event level memory. Up to 192 single event records, where each one holds the values of all inputs and date and time stamp. Storing one of these records is triggered by one input going above the event trigger level. SiViB does not store immediately when the trigger level is exceeded, but checks for the highest value during the event (peak hold).
2. Event trace memory. The event gets recorded as trace of values taken periodically. The interval time defines the time between two data records. Memory size is 192 records, where 48 of them are showing the values before trigger and the remaining 96 after the trigger. This allows to see the behavior of the values right before and after the event happened.

Event triggering

The event data sets described above will be stored when the event trigger condition is met.

Only those input channels will be used as trigger sources, which are configured as alarm triggers. Check the boxes in setup, outputs, alarm active.

As event trigger conditions, set the trigger level first. You can set different levels for alarmset 0 and 1. This allows switching the level by using digital input “Set1” (terminal 8).

The alarm delay C can be used to delay the event recording.

Upload event records to the PC

Event records can get uploaded to the PC using the serial port or the memory card. For uploading serial, click **Read Event Table** from the main menu screen.

Whenever you upload data using a memory card, the event records are uploaded as well.

Reviewing the Event Data

In the main screen click **Event Table** to show a table of the event records.

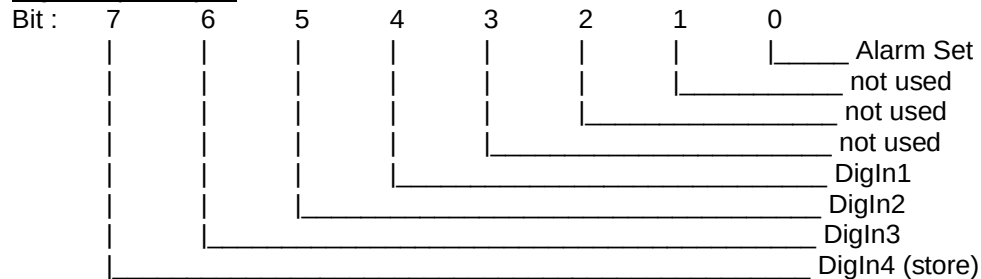
Click **Event Trend** to get a trend graph showing the traced values recorded during an event. You might need to use the zoom function of the graph to expand the time of interest.

Digital inputs – Status in Trend table

In every stored data record, the status of the digital input byte is included. This Byte shows the status of the digital inputs at the moment of storage.

Each input is represented by a specific bit according to the following table:

Digital-Inputs-Byte:



By adding the bit values, you get a decimal number describing the status of the bits.

Table of Bit Values:

Bit	Bit Value	Function
0	1	Grenzwertsatz
1	2	not used
2	4	not used
3	8	not used
4	16	DigIn1 (Event Enable)
5	32	DigIn2 (Elapsed Time Counter)
6	64	DigIn3 (Tool Change Counter)
7	128	DigIn4 (store)

Example:

DigIn2 and DigIn4 set, all other Bits 0:

$$32 + 128 = 160$$

Elapsed Time Counter, Tool Change Counter

Elapsed Time Counter

SiViB Record has an elapsed time counter, which can be used e.g. for spindle operating hours. Applying a high level (24 Volts) to Digital Input 2 (terminal 25) causes SiViB Record to count an internal clock counter in steps of 1 second. The counter value is stored in a battery backup memory. The counter can be read out using Record Control software and is shown in the bargraph window. Using the setup window, the counter can be reset to 0.

Tool Change Counter

SiViB Record has a tool change counter.

For each tool change, a pulse of minimum 200 ms duration has to be set on digital input 3 (terminal 26).

SiViB Record counts the number of valid pulses and stores them in a battery backup memory.

The counter can be read out using Record Control software and is shown in the bargraph window.

Using the setup window, the counter can be reset to 0.

Activation of the counters

The functions elapsed time counter and tool change counter have to be activated first using a key in the file Settings.ini. In the chapter [Various] use key `Counters_active=TRUE`. The instrument's counters are always on. Therefore reset them to 0 after activation.

Resetting the Counters:

Reset of the counters is done in the window Instrument Setup. Use the tab "Counters". There is a password protection preventing unauthorized changes.

Use the same password as for the other protected options.

Bargraph window

The functions elapsed time counter and tool change counter need firmware 2.9 and Record Control Software Version 3.0.14 or higher.



The window type Bargraf shows all values of one instrument in a bargraf or "tank type" format. The bars change colors according to the alarm status. The alarm setpoints are shown as markers at the side of the bar. Full scale of the bar is the selected measurement range of the individual channel. For not used channels the bar will be invisible, if the input is set to not used.

It is possible to open the bargraph window automatically on system startup, if required.

Frequency Spectras

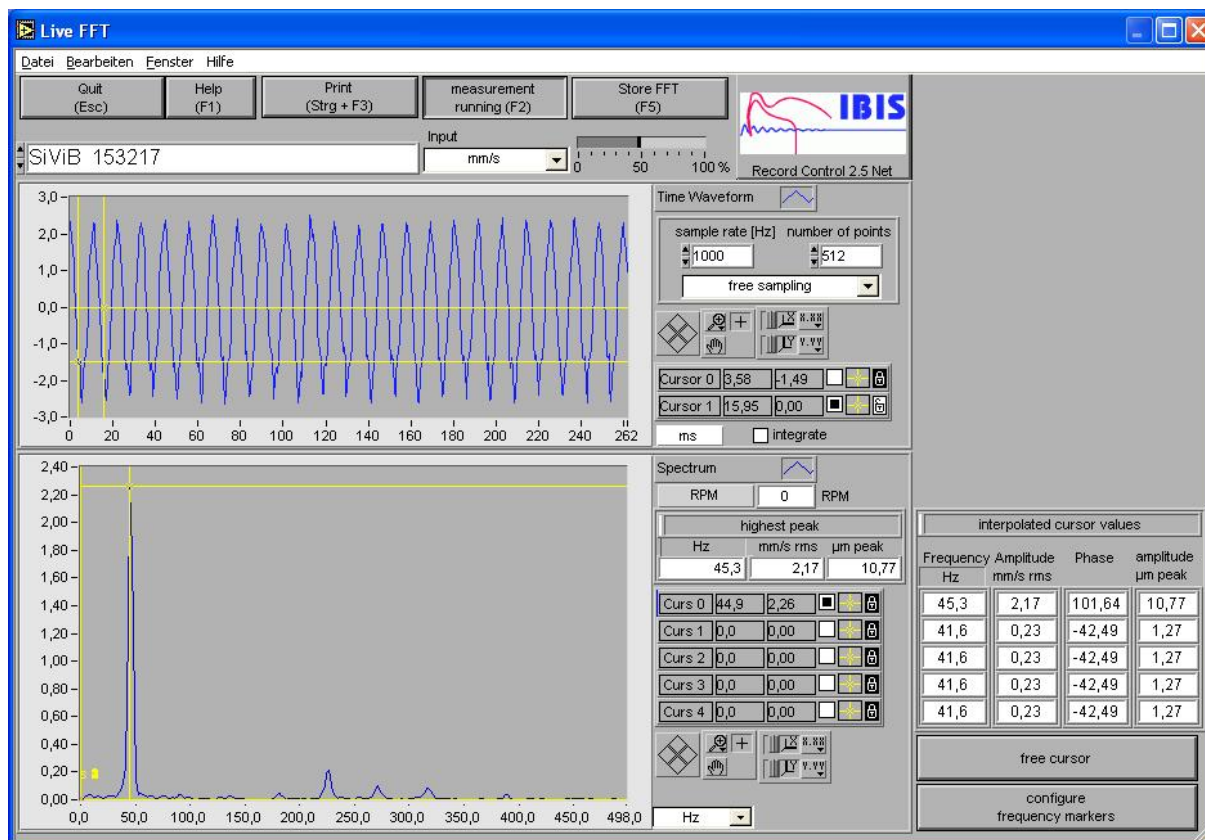
The software version Record Control Pro allows measurement and calculation of time waveforms and FFT spectras. These options are activated in Soft Control Pro only.

The spectras can be uploaded via memory card (last spectrum) or using a serial link cable. (last spectrum or live spectrum)

Live Spectrum

Connect the PC to the instrument using a nullmodem cable.

From main screen click **Live Display, Live Spectrum**.



The data acquisition and upload starts immediately. The software shows the time waveform and the FFT spectrum as soon as the transfer is completed. The next measurement is then started automatically.

The button **Measurement running / Measurement stopped** allows you to stop and freeze the last reading.

Use the controls **sample rate** and **number of points** to set the parameters according to your needs. You have 5 cursors for identifying frequency and amplitude of interesting lines. A separate table shows interpolated frequency and amplitude at the cursor positions after interpolation, which can be more precise for identifying bearing fault frequencies.

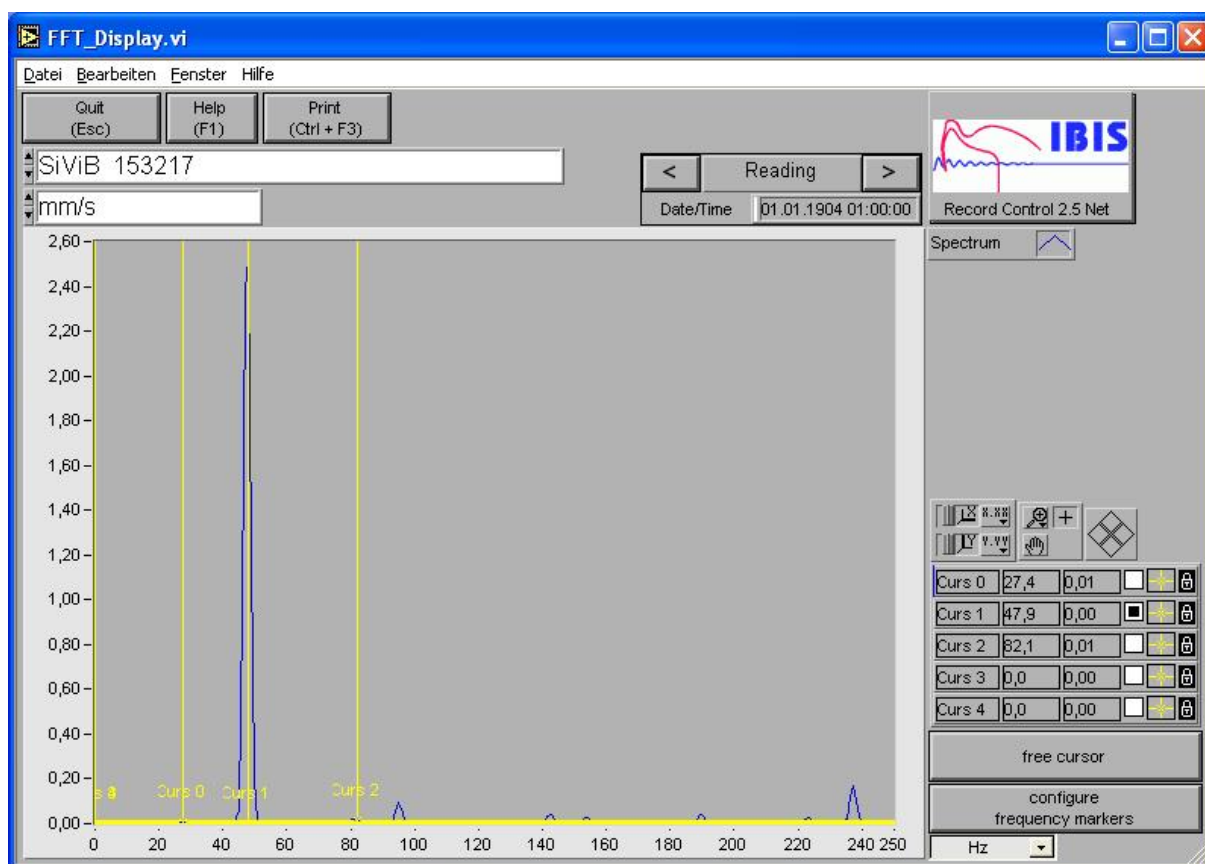
If you have a machine speed sensor connected, you can select triggered measurement for getting a phase reading. Position cursor 0 at the peak of interest and read the corresponding phase.

You can store a FFT by clicking the "Store FFT" button. These FFTs can be reviewed in the Spectrum Display window, which you can open from the main window.

The last spectrum taken in time intervals or together with overalls, can be uploaded using **Read last Spectrum serial** from main screen. Review these spectras using **Spectrum Display** from main screen.

The storage interval and sample frequency for interval spectras is set in the setup screen, Events tab. The instrument has memory for spectras as shown in the table below:

Instrument type	Memory for	Units
1501, 1511	Input accelerometer	V _{RMS} [mm/s]
1502, 1512	Input accelerometer 1 and 2	V _{RMS} [mm/s]
1503, 1513, 1603	Input accelerometer 1, 2 and 3	V _{RMS} [mm/s]
1532	Input eddy current probe 1 and 2 and accelerometer	S _{peak} [μm] V _{RMS} [mm/s]

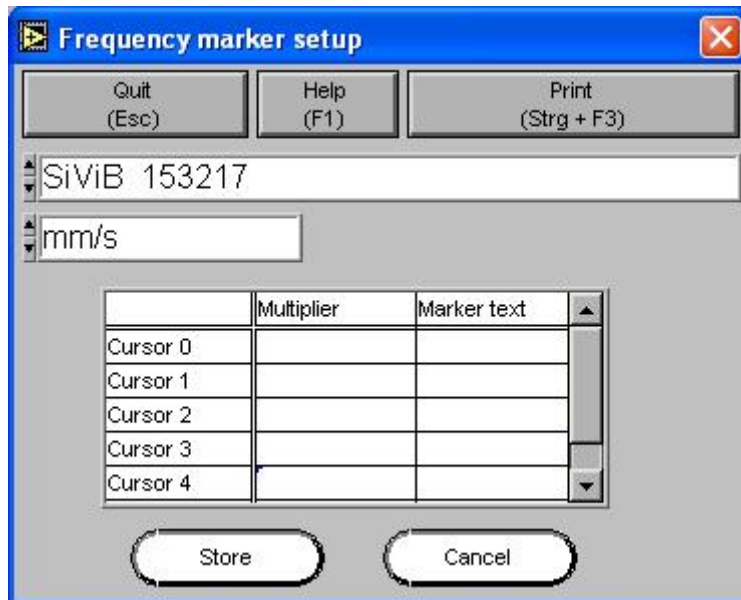


The control below the instrument selection control selects the input channel.

The arrow buttons left and right of **Measurement** select readings taken earlier or later at the same point.

Using the control right beside the lower right corner of the FFT graph you can set the frequency scaling to Hz, CPM or orders.

Use the cursors for readout of frequencies and amplitudes of interesting signal components. "Free cursor" gives you 5 independent cursor lines. Clicking on this button switches to "bearing frequency cursor". In the window "frequency marker setup" you can set multipliers and labels for the bearing frequency markers.



Usually cursor 0 is used for the machine speed, so set it to multiplier = 1.

Please note that the bearing frequencies are separate for each dynamic input of the instrument.

Export of Time Waveforms or Frequency Spectras

Time Waveform (TWF) or Frequency Spectras (FFT) can be exported to file in ASCII plain text format. This allows to use the data for further examination using other programs, e.g. Excel. Any text typed in the field „Comments“ is being stored at the beginning of the ASCII file. Name and path of the file can be entered manually, or the default name, created from date and time, can be used.

The structure of the export-file is:

The fields are Tab separated.

Field content	Explanation
Comment	Any ASCII text, no length limitation
Return	
---	3 hyphens as end of comment mark
Return	
Seconds	Date/Time in seconds since 01.01.1904
Status	Reserved for future options
Delta f	Distance of the frequency lines in Hz
n	Number of Frequency lines
Value 1	Amplitude values of each frequency line
..	
Value n	

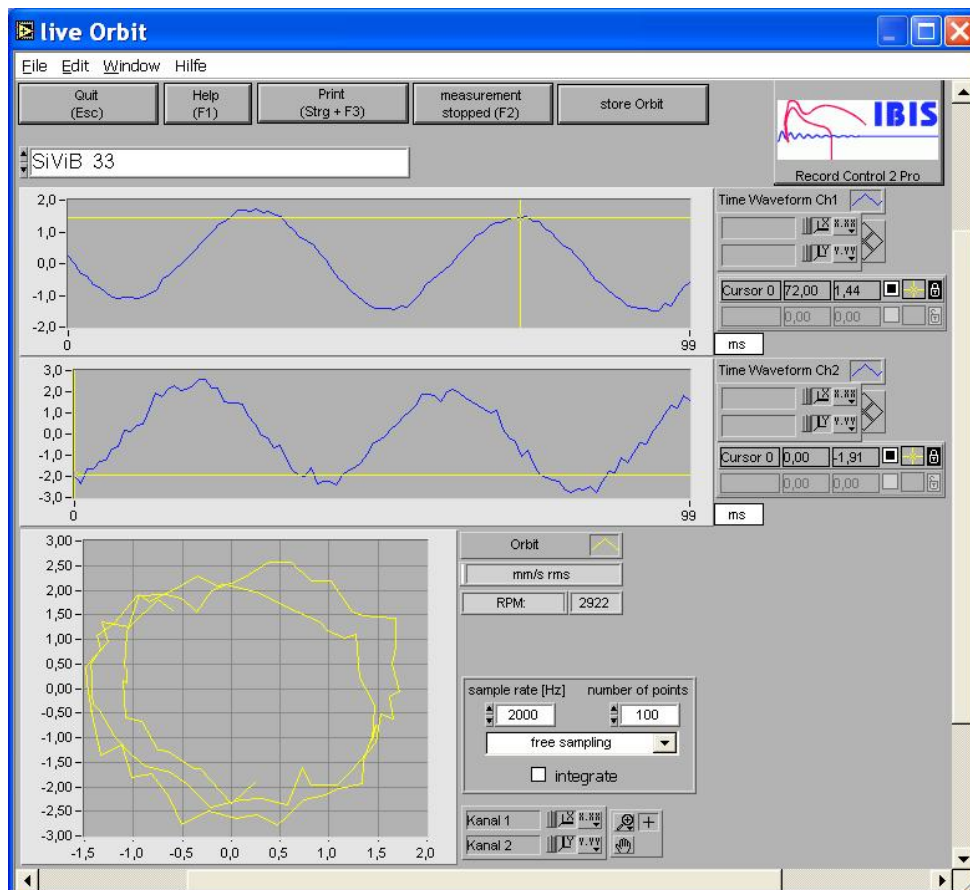
Exported TWFs and FFTs can be opened and reviewed using Record Control. Use „show exported FFT“ in the main screen.

Orbit

Record Control Pro has an orbit display for showing a XY plot for the two input signals.

You need to have:

- Software Version Record Control Pro or higher
- An instrument with 2 dynamic inputs (Typ 1502, 1512 oder 1532)
- The pickups should be oriented in XY position (perpendicular to each other)
- A machine speed reference signal (one pulse per rev)
- You must use a serial connection between instrument and PC



The orbit function samples the input signals from both inputs and uploads the data to the PC.

The upper part of the window shows the time waveform graphs. If you click on "Time Waveform" you can switch to FFT display for that channel.

The lower part of the window shows the "Orbit" graph for the input signals. Channel 1 is the X-axis and channel 2 the Y-axis.

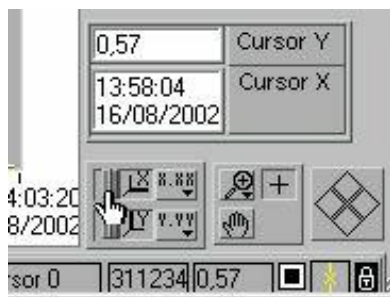
Set the sample rate and number of sample points according to your needs.

Remark: For machine speed = 0, or without speed reference pickup, triggered measurements are not possible. The control is set back to "free run" automatically.

Functions of the Graphical Plots

The graphical plots have several options allowing better inspection of the curves or customizing the plots.

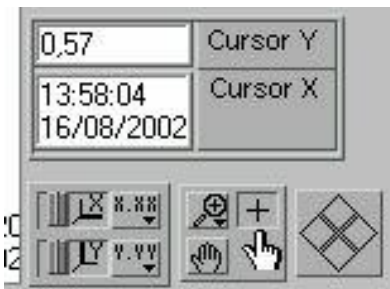
Auto Scaling



Usually autoscaling the plots is the most practical setting. If however you want to zoom in to a specific part of the graph, you must turn off auto scaling. If the sliding switch is turned to right (see picture) autoscaling is on. Click on the switch to turn it off.

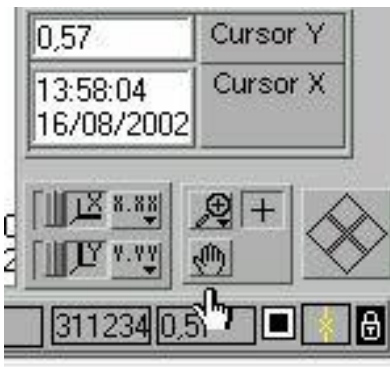
When autoscaling is off, you can set the scale of the axes manually. Click on the scale and type in the required value.

Pointing Tool



This tool is the most used in normal operation. Use it to point and click, whenever you do not want any other special functions. Select it for operating buttons or moving cursor lines.

Drag Tool



Use this tool to drag the plot. Grab the object holding the left mouse button down and drag by moving the mouse.

Zoom Tool

Hint: Zoom will work only with autoscaling off!



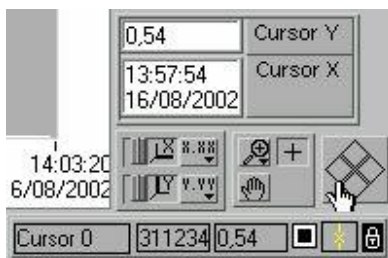
Choose the required type of zoom in the selection window.



Move the cursor to the drawing area and drag a rubber band rectangle by holding the left mouse button. When you release the mouse button, the area inside the rectangular will zoom in.

The easiest way for zooming back out is switching back to autoscaling.

Cursor Movement

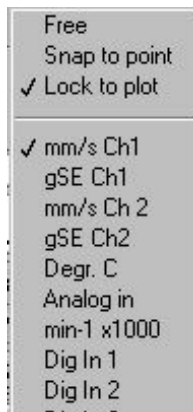


You can drag the cursor using the mouse or move it by clicking on one of the cursor movement buttons. Choose the button according to the required direction.

Cursor Curve Selection



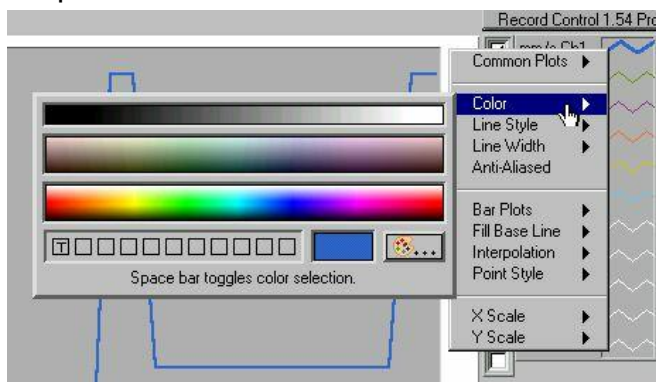
This tool selects the curve the cursor refers to for the digital display of cursor values.



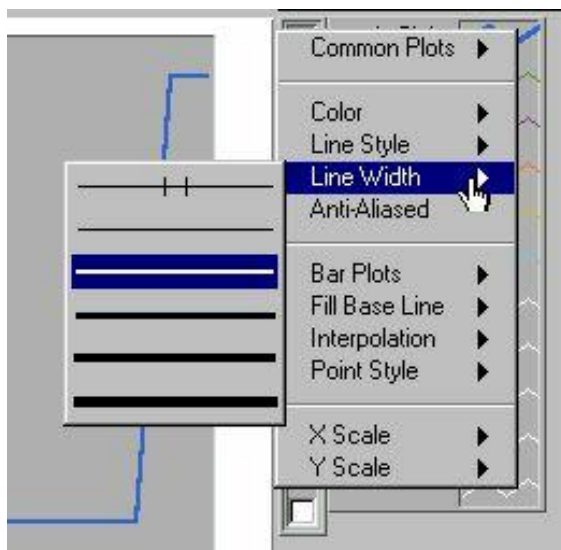
Free: Cursor can be placed everywhere. Digital display shows cursor position. Snap to point or curve: Cursor moves only from point to point along the curve.

Selection of the curve

Graph Colors



If you click on a line of the legend, the curve options window will open. Here you can select different attributes for the curves like line color, style and width.



Attention: We recommend to leave the option **Fill Base Line** to **none**, as otherwise other lines could get masked and seem to disappear.

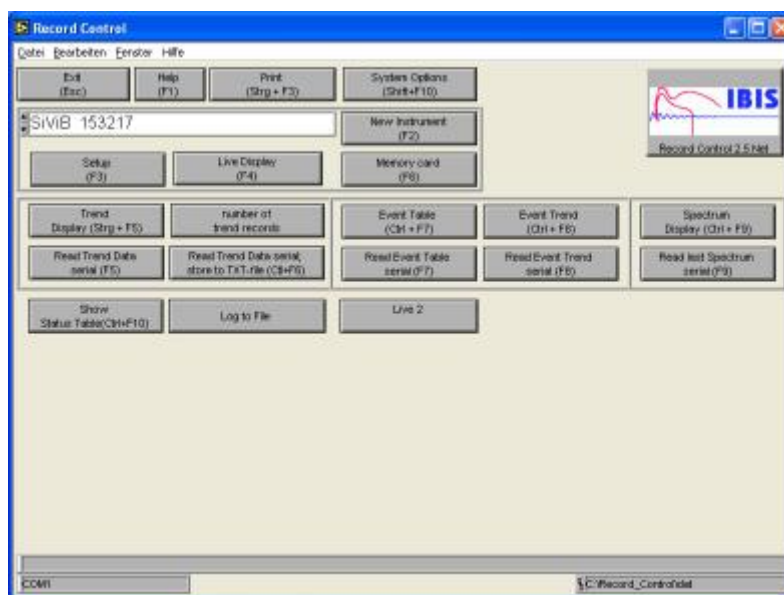
Record Control Net

Record Control Net is the software version for automatic online operation of up to 250 SiViB Record instruments networked to a PC. A CAN-Bus network or Ethernet (with additional bridges) can be used. The instruments must have the CAN-Bus option or an RS232 to Ethernet bridge fitted.

In a networked system all SiViB instruments are working independently and in the same manner as in a stand alone configuration. Only for configuration changes, data upload and live display features the communication to the PC gets active.

The Record Control Net software from the PC polls the instruments in a configurable time interval for data upload. After the upload is completed, the status display table is updated. It shows the last readings and alarm condition. The software can send alarm notification via email.

The main window has an additional button **Show Status Tables**, which gets you to the overview table



Overview Table

Meßstelle	K1	Einheit	K2	Einheit	K3	Einheit	K4	Einheit	K5	Einheit	K6	Einheit	K7 (DZ)	Einheit	Datum/Zeit
SWIB 150249	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150250	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150251	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150252	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150253	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150254	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150255	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150256	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150257	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150258	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150259	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150260	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150261	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150262	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150263	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150264	0,037	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150265	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150266	0,012	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150267	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150268	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150269	0,012	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:04
SWIB 150270	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:11
SWIB 150271	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:11
SWIB 150272	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:11
SWIB 150273	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:11
SWIB 150274	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:11
SWIB 150275	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:11
SWIB 150276	0,000	mm/s	0,000	gSE	0,000	mm/s	0,000	gSE	212,726	Grad C	212,726	Grad C	0,000	min-1x1000	28.05.2003 16:11

The overview table shows the last readings for all measurement points. Besides the numeric value, the color of the fields indicate the alarm state as follows:

Green: O.K.
 Yellow: Alarm 1
 Red: Alarm 2
 Blue: Event Alarm
 Grey: Point inactive

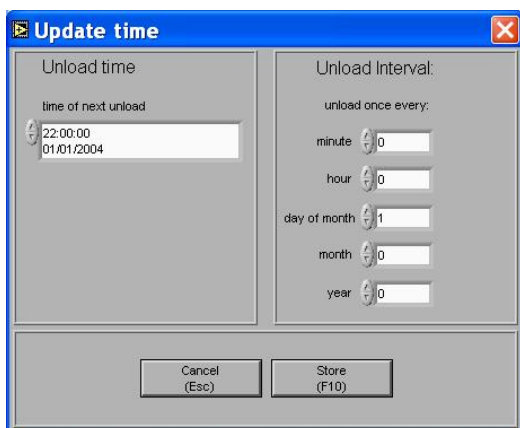
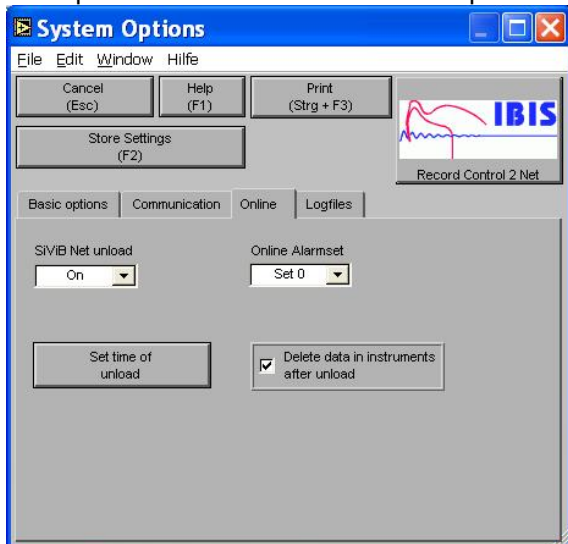
The overview table gets update whenever you open it. Clicking the button "update table" will start a table refresh using the latest uploaded data set.

Clicking the Download button manually starts a download of the trend data from the instruments to the PC.

If the checkbox "alarm mails after download" is activated, email(s) are sent, notifying of alarm conditions.

Setting Update Rate and Time

The update rate and time for the next update is set in the systems settings tab online.



Please note that all changes have to get stored.

Mail on Alarm

Record Control Net can send alarm notifications by email. After each upload process the software checks for alarm conditions and creates a alarm message file. These file is sent to all mail addresses set in the file \Dat\mailaddresses.txt

You can open and modify this file using a word processor like Notepad. There is one line per mail address and tabs between the colons.

Active	Adress	Mailserver (Proxy)	Sender adress	Account sender	Password sender
1	reciver@somewhere.com	smtpMailserver Adresse	sender@elsewere.com	Account_name_sender	Password_sender_account

There must be an active account for the sender. Otherwise the mailserver will not accept the mail. Please ask your mail system admin for advice.

The colon active can be set to 0 for disabling mails to the address in this row temporarily.

Configuration of Network connections and addresses

Set the CAN-Bus address of the instrument

In the Setup window, click "set CAN ID". After the password check you will see the CAN parameters window.



Set the CAN ID to a number between 1 and 255 unique for each instrument.

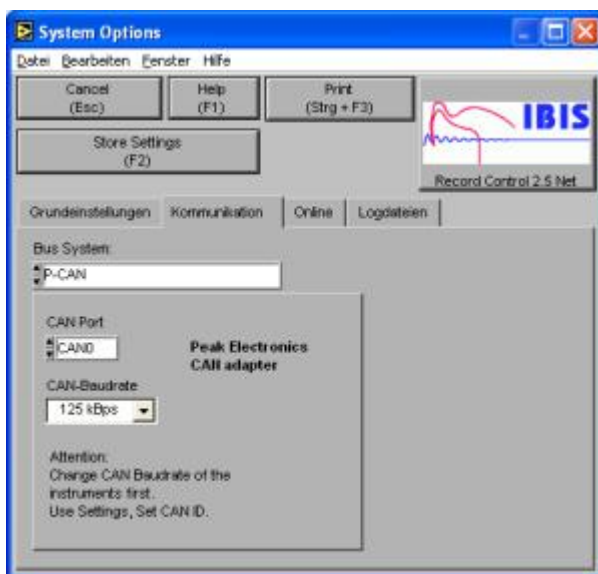
The CAN Baudrate must be the same for all instruments on the same Bus cable. 125kbps is a good choice for cables of up to 400 m length.

Please remember that a 120 ohms termination resistor must be connected to both ends of the bus cable.

You can review the CAN-Bus Ids in the table "new instrument", accessible from the main window.

You can change the CAN parameters of the instruments over CAN bus connection. All changes of CAN parameters in the instrument get effective after a reboot (power cycle) of the instrument.

The CAN parameters for the PC are set in the System Options window, tab P-CAN



Ethernet/CAN Gateway

Connections

Power supply 24 V DC

CAN-Bus cable between SiViB(s) and the gateway. Use either the sub D 9-pin connector or the screw terminals.

Termination resistor at both ends of the CAN-Bus cable. If the gateway is on one cable end, you can use the internal termination resistor by switching the corresponding dip switch to on.

Ethernet cable from the gateway to a hub or switch of your Ethernet system.

Configuration of the gateway

The gateway needs a fix IP address. Further, you will need to know the Subnet mask of your network system.

You can set the address and other options for the gateway by editing the file EthCANGateway.txt in the record Control software folder. Open the file using a text editor, e.g. Edit or Wordpad. When you have stored the changes, the file has to be transferred to the gateway.

Connect the PC to the gateway using the RS232 connectors and a nullmodem cable.

Start Hyperterminal by double clicking Eth_CAN.ht in the record control folder.

You will get a message like this:

```
Welcome at SYSTEC CAN-Ethernet Gateway V1.0.758
-----type the following comands as shown below
/ > cd save
/save > rm rc
/save > ls
..          typ:0xc2
/save > write rc
-----copy the content of the file EthCANGateway.txt to here
siocfg 9600
ipcfg 192.168.42.92 255.255.255.0 192.168.42.1
mkif led
/if/led0/fin +l +hd1
mkif can
mkif btpserv
mkif tcperv
/if/can0 baud:4 canid:0 buf:500 on
-----type Ctrl-D for stopping the write mod and type:
/save > ls
..          typ:0xc2
rc          use:0 typ:0x20 len:159
/save > sync
/save >
```

Close Hyperterminal and boot the gateway.

Configure the Software

Go to the system options screen, tab communication, select "Ethernet-CAN Gateway. Enter the Ethernet adress of the gateway. Leave the port set to the standard setting "8234". Store the changes and close and reopen the software.

Signal LEDs on the gateway

Power: Power supply is O.K.

Link: connetion to the Ethernet hub is O.K.

Connect: Connection to Record Control software is established. Will go off some seconds after closing the software.

Active and Traffic are blinking when there is traffic on the system

Error Ethernet and Error CAN go on or blinking when there are errors in the communication.

Annex

Technical Specification

Specifications

Units of Measurement:	1, 2 or 3 x vibration velocity v_{RMS} and roller bearing condition gSE 4 or 2 x process parameters (0-10 Volt) or temperature (PT1000, KTY 84) 3 x digital inputs 24 VDC 1 x machine speed (1 pulse per rev)
Case:	Plastic case IP20 for 35 mm DIN rail or base plate mounting Dimensions 100 x 75 x 110 mm (WxHxD);
Inputs / Outputs:	30 screw terminals: 1 or 2 inputs for accelerometers; ICP supply sensitivity 10, 100 or 500 mV/g 4 or 2 inputs for DC Volts (0 – 10 Volts) or temperature (PT1000, KTY 84) 1 input for machine status / parameter set (Opto coupler) 1 input for trigger signal store data set 3 digital inputs 24 VDC (opto coupler) 3 Alarm outputs (opto coupler) 1 output sensor fault (opto coupler)
Serial interface RS232C:	9-pin male; for Null-Modem-Cable to PC for configuration and download
Power supply:	24 VDC; approx. 250 mA
Ranges: (using 100 mV/g accelerometer)	vibration: 0,1 - 10 / 20 / 40 / 80 mm/s v_{eff} bearing condition: 0,1 - 10 / 20 / 40 / 80 gSE process variables: 0 – 10 VDC temperature: 0 – 200 degrees Celsius speed: 0 – 60 000 min ⁻¹ (1 pulse per rev)

Types and part numbers:

Unit	Description	Order Number
SiViB Record 1501	1 input for accelerometer 100 mV/g, 4 inputs for temperature or 0 – 10 VDC	SiV01.011
SiViB Record 1502	2 input for accelerometers 100 mV/g, 2 inputs for temperature or 0 – 10 VDC	SiV01.012
SiViB Record 1503	3 input for accelerometers 100 mV/g	SiV01.013
SiViB Record 1511	1 input for accelerometer 10 mV/g, 4 inputs for temperature or 0 – 10 VDC	SiV01.021
SiViB Record 1512	2 input for accelerometers 10 mV/g, 2 inputs for temperature or 0 – 10 VDC	SiV01.022
	3 inputs for accelerometers 10 mV/g	SiV01.023
SiViB Record WS 1532	1 Input for accelerometer 100 mV/g 2 inputs for eddy current probes 8 mV/ μ m	SiV01.032
SiViB Record Control Light	Configuration software for Windows (free with instrument)	SiV09.030
SiViB Record Control Pro	For configuration and advanced diagnosis functions	SiV09.031
SiViB Record Control Net	For configuration, analysis and network functions	SiV09.032
Nullmodem cable	Serial cable PC – SiViB for configuration and download 9-pin socket – 9-pin socket	SiV09.040
Memory card	Memory card (Smartcard) for transfer of data records and configuration between SiViB Record and PC	SiV09.041
Card reader	Card reader for PC (RS232 + PS/s)	SiV09.042
Display	External display for SiViB Record	SiV09.902

Record Control Software Versions

Function	Light	Pro	Net
Configure Ranges	•	•	•
Configure Alarms	•	•	•
Set SiViB Clock	•	•	•
Upload Measurement Data	•	•	•
Communication via RS232C	•	•	•
Trendgraph	•	•	•
Trend Table	•	•	•
Configure Event Alarm	•	•	•
Event Table	•	•	•
Event Trend	•	•	•
Live-Display for Overalls	•	•	•
Upload and display Spectras	-	•	•
Live Spectras	-	•	•
Live Time Waverform	-	•	•
Live Phase	-	•	•
Orbit-Display	-	•	•
Online Functions via bus system	-	-	•
Status-Overview	-	-	•
Mail on Alarm	-	-	•

Frequently Asked Questions

Question: When I start the software I get the following error message:

Error 37 initializing COM port
Device not found

Answer 1: Maybe you have your chip card reader connected to the serial port which is configured for serial communication to the instrument. If you confirm by clicking O.K. you can use all functions of the software except those which are using the serial port. You could consider to move the card reader to a different port for avoiding the message.

Answer 2: Maybe a 3rd party software is using the serial port of your PC. Stop this software or configure a different port to be used.

Question: Can I use a card reader from an other manufacturer?

Answer:

The Record Control Software uses drivers for the card reader which are instrument specific. Therefore it is not possible to use other card readers presently.

Question:

On my SiViB the red LED „Card Error“ is on, even without a card plugged in.

Answer:

If the configuration in the internal memory of SiViB gets corrupted, the red LED „Card Error“ will go on. Please check the configuration setup and send it to the instrument. As soon as SiViB has a valid setup, the red LED should go off.

Question:

What is the difference between ***Read Trend Data serial*** and ***Read Trend Data serial store to TXT file***?

Answer:

Read Trend Data serial stores the data to the trend data files. The data from these files is used when displaying a trend graph. Data uploaded at a later time is appended to the same file and the trend graph updated accordingly. The file names for these files are set by Record Control.

Read Trend Data and store to TXT file should be used if you want to open the file with a spreadsheet program for further calculations or customized graphs. You will be prompted for a file name. These files can not be used by Record Control for showing trend graphs.